SECOND-ORDER HERMITE–BIEHLER DE BRANGES KERNELS, THE DERIVATIVE-WRONSKIAN SPLITTING $K_{\alpha,\beta} \equiv \frac{\alpha}{\pi}\widetilde{W}_{0,1} + \frac{\alpha\beta}{\pi}\widetilde{W}_{1,2}$, AND $d = 0$ LIFTING OF THE LEHMER NEAR-COALESCENCE DEGENERACY FOR RIEMANN'S Ξ-FUNCTION

THIJS MISSIAEN

October 25, 2026

Abstract. We prove that for the second-order Hermite–Biehler structure function $E_{\alpha,\beta}(z) := \Xi(z) - \mathrm{i}\alpha\Xi'(z) - \beta\Xi''(z)$ ($\alpha > 0, \beta > 0$) studied by Lagarias, Ki, and Suzuki, the de Branges kernel $K_{\alpha,\beta}(t - \mathrm{i}d, t - \mathrm{i}d)$ exhibits an exact cancellation of all three squared moduli ($|\Xi|^2, \alpha^2|\Xi'|^2, \beta^2|\Xi''|^2$) and of the $(0,2)$ cross-term $-2\beta\mathrm{Re}(\overline{\Xi}\Xi'') \equiv 0$, splitting into $K_{\alpha,\beta}(t - \mathrm{i}d, t - \mathrm{i}d) \equiv \frac{\alpha}{\pi}\widetilde{W}_{0,1}(d,t) + \frac{\alpha\beta}{\pi}\widetilde{W}_{1,2}(d,t)$, where $\widetilde{W}_{1,2}(d,t) = \frac{1}{2d}\partial_d|\Xi'(t - \mathrm{i}d)|^2$ is the parity-reduced derivative Wronskian. Consequently, as $d \to 0^+$, $\pi K_{\alpha,\beta}(t,t) \equiv \alpha c_1(t) + \alpha\beta c_1^{(1)}(t)$, so at the midpoint t_{mid} of any closely spaced Lehmer pair ($\Xi'(t_{\mathrm{mid}}) = 0$), $\pi K_{\alpha,\beta}(t_{\mathrm{mid}}, t_{\mathrm{mid}}) \equiv -\alpha\Xi(t_{\mathrm{mid}})\Xi''(t_{\mathrm{mid}}) + \alpha\beta(\Xi''(t_{\mathrm{mid}}))^2 \to +\alpha\beta(\Xi''(t_0))^2 > 0$ remains strictly positive on $d = 0$ at order d^0. Finally, we prove the radial four-dimensional Laplace coupling $\widetilde{W}_{1,2} \equiv \frac{1}{4}(\partial_d^2 + \frac{2}{d}\partial_d + \partial_t^2)\widetilde{W}_{0,1}$, the dipole recursion $c_{2m-1}^{(1)}(t) \equiv \frac{1}{4}(2m(2m+1)c_{2m+1}(t) + c_{2m-1}''(t))$, and the order-$M$ Hurwitz coalescence law $\pi K_{\boldsymbol{\alpha}}(t_0,t_0) = +\alpha_{m-1}\alpha_m(\Xi^{(m)}(t_0))^2 + \mathcal{O}(\alpha_{>m}) > 0$.

1. Introduction and Main Results

In the analytic theory of Hilbert spaces of entire functions [3, 4, 5] associated with Riemann's completed zeta function $\Xi(z) := \xi(\frac{1}{2} + \mathrm{i}z)$ ($z = t - \mathrm{i}d$, $s = \frac{1}{2} + d + \mathrm{i}t$) [25, 24, 29], a central role is played by canonical Hermite–Biehler structure functions obtained by applying differential polynomials to $\Xi(z)$ [12, 10, 28]. While Conrey and Li [1] demonstrated that global inner-product positivity conditions on classical Sonine–de Branges spaces fail for $\xi(s)$, Lagarias [11, 12], Ki [10], and Suzuki [28] investigated the direct Hermite–Biehler hierarchy generated by linear, quadratic, and higher-order differential operators:

$$E_\alpha(z) := \Xi(z) - \mathrm{i}\alpha\,\Xi'(z), \qquad E_{\alpha,\beta}(z) := \Xi(z) - \mathrm{i}\alpha\,\Xi'(z) - \beta\,\Xi''(z) \qquad (\alpha > 0,\ \beta > 0), \tag{1.1}$$

including the canonical repeated-root family $E_{\alpha,\alpha^2/4}(z) = (1 - \frac{\mathrm{i}\alpha}{2}\partial_z)^2\Xi(z)$ ($\beta = \alpha^2/4$).

In Paper IV of this series [17] (building on Papers I–III [14, 15, 16]), we proved that for the first-order family ($\beta = 0$), the de Branges reproducing kernel on the diagonal $w = t - \mathrm{i}d$ collapses to $K_\alpha(w,w) \equiv \frac{\alpha}{\pi}\widetilde{W}_{0,1}(d,t)$, where $\widetilde{W}_{0,1}(d,t) := \frac{1}{2d}\partial_d|\Xi(t - \mathrm{i}d)|^2 = d^{-1}\mathrm{Im}(\overline{\Xi(w)}\Xi'(w))$ is the parity-reduced transverse Wronskian [27, 8, 2, 7]. Although this identity decouples local off-critical zero exclusion from the global Sonine obstructions of Conrey and Li [1], the first-order kernel $K_\alpha(w,w)$ exhibits a structural degeneracy on the critical line $d = 0$ itself whenever two zeros approach coalescence [13]: at a hypothetical double zero $t_0 \in \mathbb{R}$ ($\Xi(t_0) = \Xi'(t_0) = 0$, $\Xi''(t_0) \neq 0$), the critical-line value $\pi K_\alpha(t_0,t_0) = \alpha((\Xi'(t_0))^2 - \Xi(t_0)\Xi''(t_0)) = \alpha c_1(t_0) = 0$ vanishes at order d^0, so strict

Date: October 25, 2026.

2020 *Mathematics Subject Classification.* Primary 11M06, 11M26; Secondary 30D15, 46E22.

Key words and phrases. Riemann Ξ-function, transverse Wronskian, de Branges reproducing kernel, Hermite–Biehler polynomials, Lucas–Jensen derivative Wronskian, Lehmer pairs, Hurwitz polynomials, Wigner phase-space distribution, radial Laplace PDE.

ORCID: 0009-0007-3678-0810.

positivity emerges only transversely at order d^2 via $c_3(t_0) = \frac{1}{2}(\Xi''(t_0))^2 > 0$. Equivalently, at the critical-line extremum t_{mid} ($\Xi'(t_{\mathrm{mid}}) = 0$) between a closely spaced Lehmer pair $t_\pm = t_{\mathrm{mid}} \pm \delta$ ($\delta \to 0$), the first-order barrier $\pi K_\alpha(t_{\mathrm{mid}}, t_{\mathrm{mid}}) = -\alpha \Xi(t_{\mathrm{mid}})\Xi''(t_{\mathrm{mid}}) = \mathcal{O}(\delta^2) \to 0$ becomes arbitrarily small on $d = 0$.

The purpose of the present paper (Paper XIII, continuing Papers I–X [14, 15, 16, 17, 18, 19, 20, 21, 22, 23]) is to prove that passing to the second-order Lagarias–Ki–Suzuki structure function $E_{\alpha,\beta}(z)$ ($\beta > 0$) resolves this $d = 0$ near-coalescence degeneracy completely at order d^0, without introducing any cross-term contamination into the reproducing kernel. Our main results are consolidated in the following theorem.

Theorem 1.1 (Second-Order De Branges Splitting, $d = 0$ Lehmer Lifting, Radial Laplace Recursion, and Order-M Hurwitz Chains)**.** *Let $z = t - \mathrm{i}d \in \mathbb{C}$ with $d \in \mathbb{R} \setminus \{0\}$ and $t \in \mathbb{R}$, and let $E_{\alpha,\beta}(z) := \Xi(z) - \mathrm{i}\alpha\Xi'(z) - \beta\Xi''(z)$ and $E^{\#}_{\alpha,\beta}(z) := \overline{E_{\alpha,\beta}(\overline{z})} = \Xi(z) + \mathrm{i}\alpha\Xi'(z) - \beta\Xi''(z)$ for $\alpha > 0, \beta > 0$. Then:*

(i) ***Exact Cross-Term-Free Second-Order De Branges Splitting and Cauchy–Pick Decomposition.*** *In the numerator $|E_{\alpha,\beta}(t - \mathrm{i}d)|^2 - |E^{\#}_{\alpha,\beta}(t - \mathrm{i}d)|^2$, all three quadratic norms $|\Xi|^2$, $\alpha^2|\Xi'|^2$, and $\beta^2|\Xi''|^2$ as well as the $(0,2)$ cross-term $-2\beta\mathrm{Re}(\overline{\Xi}\Xi'') \equiv 0$ cancel identically, yielding the exact global decomposition*

$$K_{\alpha,\beta}(t - \mathrm{i}d,\, t - \mathrm{i}d) := \frac{|E_{\alpha,\beta}(t - \mathrm{i}d)|^2 - |E^{\#}_{\alpha,\beta}(t - \mathrm{i}d)|^2}{4\pi d} \equiv \frac{\alpha}{\pi}\,\widetilde{W}_{0,1}(d,t) + \frac{\alpha\beta}{\pi}\,\widetilde{W}_{1,2}(d,t), \tag{1.2}$$

where $\widetilde{W}_{k,k+1}(d,t) := d^{-1}\mathrm{Im}(\overline{\Xi^{(k)}(t - \mathrm{i}d)}\Xi^{(k+1)}(t - \mathrm{i}d)) = \frac{1}{2d}\partial_d|\Xi^{(k)}(t - \mathrm{i}d)|^2$ ($k = 0, 1$). Moreover, for any finite set of nodes $z_1, \dots, z_M \in \mathbb{C}_-$, the $M \times M$ Cauchy–Pick Gram matrix $[\mathbf{G}_{\alpha,\beta}]_{j,k} = K_{\alpha,\beta}(z_j, z_k)$ splits identically as $\mathbf{G}_{\alpha,\beta} \equiv \alpha\mathbf{G}^{(0,1)} + \alpha\beta\mathbf{G}^{(1,2)}$ with zero non-adjacent jump terms.

(ii) ***Lifting the $d = 0$ Lehmer Near-Coalescence Degeneracy at Order d^0 and Square Interlocking.*** *Taking the limit $d \to 0^+$ in (1.2) yields the exact critical-line identity*

$$\pi\, K_{\alpha,\beta}(t,t) \equiv \alpha\Big((\Xi'(t))^2 - \Xi(t)\Xi''(t)\Big) + \alpha\beta\Big((\Xi''(t))^2 - \Xi'(t)\Xi'''(t)\Big) = \alpha\, c_1(t) + \alpha\beta\, c_1^{(1)}(t). \tag{1.3}$$

At any critical-line extremum t_{mid} ($\Xi'(t_{\mathrm{mid}}) = 0$) between a closely spaced Lehmer pair $t_\pm = t_{\mathrm{mid}} \pm \delta$ ($\delta > 0$), the product $-\Xi'(t_{\mathrm{mid}})\Xi'''(t_{\mathrm{mid}}) \equiv 0$ vanishes identically, so $\pi K_{\alpha,\beta}(t_{\mathrm{mid}}, t_{\mathrm{mid}}) \equiv -\alpha\Xi(t_{\mathrm{mid}})\Xi''(t_{\mathrm{mid}}) + \alpha\beta(\Xi''(t_{\mathrm{mid}}))^2 \xrightarrow{\delta\to 0} +\alpha\beta(\Xi''(t_0))^2 > 0$ remains strictly positive on $d = 0$ itself at order d^0. Globally along $d = 0$, pure squares interlock across zeros γ_k ($c_1(\gamma_k) = (\Xi'(\gamma_k))^2 > 0$), extrema τ_k ($c_1^{(1)}(\tau_k) = (\Xi''(\tau_k))^2 > 0$), and inflection points ν_k ($c_1(\nu_k) = (\Xi'(\nu_k))^2 > 0$).

(iii) ***Radial 4D-Laplace Coupling, Universal Dipole Recursion, and Order-M Hurwitz Chains.*** *The derivative Wronskian $\widetilde{W}_{1,2}(d,t)$ is coupled to $\widetilde{W}_{0,1}(d,t)$ by the exact radial four-dimensional Laplace operator and Taylor coefficient recursion*

$$\widetilde{W}_{1,2}(d,t) \equiv \frac{1}{4}\left(\frac{\partial^2\widetilde{W}_{0,1}}{\partial d^2} + \frac{2}{d}\frac{\partial\widetilde{W}_{0,1}}{\partial d} + \frac{\partial^2\widetilde{W}_{0,1}}{\partial t^2}\right), \qquad c_{2m-1}^{(1)}(t) \equiv \frac{1}{4}\Big(2m(2m+1)\,c_{2m+1}(t) + c_{2m-1}''(t)\Big) \tag{1.4}$$

for all $m \geq 1$ (in particular $c_1^{(1)}(t) \equiv \frac{3}{2}c_3(t) + \frac{1}{4}c_1''(t)$). In Wigner phase space [30, 9], $\widetilde{W}_{1,2}(d,t) = \frac{1}{2}\int_0^\infty y^2 \mathrm{sinc}_{\mathrm{hyp}}(dy) I^{(1)}(y,t)\,\mathrm{d}y$ with $I^{(1)}(y,t) = \frac{1}{4}(y^2 + \partial_t^2)I(y,t)$, which enhances the Riemann–Siegel diagonal [26, 6] to $\sim \frac{1}{80}(\ln\frac{t}{2\pi})^5$ and applies quartic $\mathcal{O}(N(t)^{-5}) = \mathcal{O}(t^{-5/2})$ self-damping to the slowest boundary mode $(N-1, N)$. More generally, for any degree-M real Hurwitz polynomial $P_M(X) = \sum_{r=0}^M \alpha_r X^r$ ($\alpha_0 = 1, \alpha_r > 0$), $E_{\boldsymbol{\alpha}}(z) := P_M(-\mathrm{i}\partial_z)\Xi(z)$ satisfies at any multiplicity-m critical-line zero t_0 ($1 \leq m \leq M$) the universal $d = 0$ pure-square law $\pi K_{\boldsymbol{\alpha}}(t_0, t_0) = +\alpha_{m-1}\alpha_m(\Xi^{(m)}(t_0))^2 + \mathcal{O}(\alpha_{>m}) > 0$.

The canonical quantitative constants governing Theorem 1.1 (for $\alpha=1,\beta=\frac{1}{4}$) are given explicitly by

$$\begin{aligned} K_{1,1/4}(\gamma_1,\gamma_1) &= 6.905897\times10^{-7}, & c_1^{(1)}(\gamma_1) &= 1.030558\times10^{-6},\\ c_1(\tau_2) &= 6.369715\times10^{-11}, & c_1^{(1)}(\tau_2) &= \big(\Xi''(\tau_2)\big)^2 = 6.741923\times10^{-11},\\ \lim_{\Delta\gamma\to0}\pi K_{1,1/4}(t_{\mathrm{mid}},t_{\mathrm{mid}}) &= 1.685481\times10^{-11}, & \frac{|C^{(1,2)}_{N-1,N}(10^8)|}{\mathcal{D}_{1,2}(10^8)} &= 3.5759\times10^{-30}. \end{aligned} \tag{1.5}$$

1.1. **Proof overview.** The proof of Theorem 1.1 combines three quantitative ingredients developed in Sections 2–4:

(1) *Hermitian polarization of even/odd derivative jets and Cauchy–Riemann cancellation (Section 2).* Decomposing $E_{\alpha,\beta}(z)=A_\beta(z)-\mathrm{i}\alpha B(z)$ and $E^{\#}_{\alpha,\beta}(z)=A_\beta(z)+\mathrm{i}\alpha B(z)$ into the real-entire even-order combination $A_\beta(z):=\Xi(z)-\beta\Xi''(z)$ and the odd-order derivative $B(z):=\Xi'(z)$, we prove that the Hermitian sum $A_\beta(z_1)\overline{A_\beta(z_2)}+\alpha^2B(z_1)\overline{B(z_2)}$ cancels identically in $E_{\alpha,\beta}(z_1)\overline{E_{\alpha,\beta}(z_2)}-E^{\#}_{\alpha,\beta}(z_1)\overline{E^{\#}_{\alpha,\beta}(z_2)}$. On the diagonal $z_1=z_2=w$, this eliminates $|\Xi|^2$, $\alpha^2|\Xi'|^2$, $\beta^2|\Xi''|^2$, and $-2\beta\mathrm{Re}(\overline{\Xi}\Xi'')\equiv0$ simultaneously, yielding (1.2).

(2) *Order-d^0 pure-square collapse at critical-line extrema and Lehmer midpoints (Section 3).* Expanding $\widetilde{W}_{1,2}(d,t)=\sum_{m=1}^{\infty}c^{(1)}_{2m-1}(t)d^{2m-2}$ in powers of d^2, we evaluate the critical-line limit (1.3). Because $\Xi'(t_{\mathrm{mid}})=0$ holds identically at the local extremum t_{mid} between any pair of simple zeros $t_\pm=t_{\mathrm{mid}}\pm\delta$, the derivative cross-term $-\Xi'(t_{\mathrm{mid}})\Xi'''(t_{\mathrm{mid}})\equiv0$ vanishes without remainder, leaving the strictly positive pure square $+\alpha\beta(\Xi''(t_{\mathrm{mid}}))^2>0$ on $d=0$ for every $\delta>0$ and in the coalescence limit $\delta\to0$.

(3) *Subharmonic Laplace commutation on $|\Xi|^2$ and Bézout–Hurwitz parity alternation (Section 4).* Using the identity $(\partial_d^2+\partial_t^2)|f(t-\mathrm{i}d)|^2\equiv4|f'(t-\mathrm{i}d)|^2$ for holomorphic functions and commuting ∂_d with $\Delta_{d,t}=\partial_d^2+\partial_t^2$, we obtain $W_{1,2}\equiv\frac{1}{4}\Delta_{d,t}W_{0,1}$, which upon substituting $W_{0,1}=d\,\widetilde{W}_{0,1}$ yields (1.4). Finally, for a general degree-M Hurwitz polynomial $P_M(X)=P_{\mathrm{even}}(X^2)+XP_{\mathrm{odd}}(X^2)$, we prove that the sign alternation $(-1)^{j+k}$ in the Bézout–Wronskian expansion always aligns constructively at a multiplicity-m zero to produce $+\alpha_{m-1}\alpha_m(\Xi^{(m)}(t_0))^2>0$.

1.2. **Notation and conventions.** Throughout this paper, $s=\sigma+\mathrm{i}t=\frac{1}{2}+d+\mathrm{i}t$ with $d=\sigma-\frac{1}{2}\in\mathbb{R}$ and $t\in\mathbb{R}$, and $z=-\mathrm{i}(s-\frac{1}{2})=t-\mathrm{i}d$ (so $z\in\mathbb{C}_-$ corresponds to $d=-\mathrm{Im}(z)>0$, while $\bar z=t+\mathrm{i}d\in\mathbb{C}_+$). Riemann's completed zeta function is $\xi(s):=\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ and $\Xi(z):=\xi(\frac{1}{2}+\mathrm{i}z)=R(d,t)+\mathrm{i}\,I(d,t)$. Because $\Xi(z)$ is real-entire ($\overline{\Xi(\bar z)}=\Xi(z)$), every derivative $\Xi^{(k)}(z):=\frac{\mathrm{d}^k}{\mathrm{d}z^k}\Xi(z)$ is also real-entire. Under $z=t-\mathrm{i}d$, the Cauchy–Riemann operator is $\partial_d\Xi^{(k)}(t-\mathrm{i}d)=-\mathrm{i}\,\Xi^{(k+1)}(t-\mathrm{i}d)$ and $\partial_d^2\Xi^{(k)}(t-\mathrm{i}d)=-\Xi^{(k+2)}(t-\mathrm{i}d)$. For each integer $k\ge0$, the k-th transverse Wronskian and its parity-reduced counterpart are defined by

$$W_{k,k+1}(d,t):=\frac{1}{2}\partial_d\big|\Xi^{(k)}(t-\mathrm{i}d)\big|^2=\mathrm{Im}\Big(\overline{\Xi^{(k)}(t-\mathrm{i}d)}\,\Xi^{(k+1)}(t-\mathrm{i}d)\Big),\quad \widetilde{W}_{k,k+1}(d,t):=\frac{W_{k,k+1}(d,t)}{d}. \tag{1.6}$$

In the logarithmic u-domain, $\Xi(z)=8\int_0^\infty\Phi(u)\cos(2zu)\,\mathrm{d}u$, where $\Phi(u):=\sum_{n=1}^{\infty}(2\pi^2n^4\mathrm{e}^{9u}-3\pi n^2\mathrm{e}^{5u})\exp(-\pi n^2\mathrm{e}^{4u})$. All high-precision numerical tables and interval enclosures are generated by the standalone verification script `scripts/verify_paper13_second_order_debranges.py` at 60-decimal (`mpmath.mp.dps = 60`), 550-decimal (`mp.workdps(550)`), and 35-decimal interval (`mpmath.iv.dps = 35`) precision.

2. The Exact Second-Order De Branges Reproducing Kernel Identity and Cauchy–Pick Gram Splitting

In the classical theory of de Branges Hilbert spaces $\mathcal{H}(E)$ [3, 4], an entire function $E(z)$ belongs to the Hermite–Biehler class (with respect to the lower half-plane $\mathbb{C}_- = \{w = t - \mathrm{i}d : d > 0\}$) if $|E(w)| > |E^{\#}(w)|$ for all $w \in \mathbb{C}_-$, where $E^{\#}(z) := \overline{E(\bar{z})}$. Equivalently, for the upper half-plane reflection $E^*(z) := E^{\#}(z)$ studied by Lagarias [12], Ki [10], and Suzuki [28], $|E^*(t + \mathrm{i}d)| = |E(t - \mathrm{i}d)| > |E^{\#}(t - \mathrm{i}d)| = |E^*(t - \mathrm{i}d)|$ for all $d > 0$. When passing from the first-order structure function $E_\alpha(z) = \Xi(z) - \mathrm{i}\alpha\Xi'(z)$ to the second-order structure function $E_{\alpha,\beta}(z) = \Xi(z) - \mathrm{i}\alpha\Xi'(z) - \beta\Xi''(z)$, a naive expansion of $|E_{\alpha,\beta}(w)|^2$ produces six real quadratic and bilinear pairings among $\Xi(w), \Xi'(w), \Xi''(w)$, including the sign-indefinite non-adjacent pairing $-2\beta\mathrm{Re}(\overline{\Xi(w)}\Xi''(w))$. Our first theorem proves that all four even-parity pairings cancel identically in the reproducing kernel.

Theorem 2.1 (The Exact Second-Order De Branges Reproducing Kernel Identity)**.** *For all $\alpha, \beta \in \mathbb{R}$ and every $w = t - \mathrm{i}d \in \mathbb{C}$ with $d \neq 0$, let $E_{\alpha,\beta}(z) := \Xi(z) - \mathrm{i}\alpha\Xi'(z) - \beta\Xi''(z)$ and $E^{\#}_{\alpha,\beta}(z) := \Xi(z) + \mathrm{i}\alpha\Xi'(z) - \beta\Xi''(z)$. Then in the difference $|E_{\alpha,\beta}(t - \mathrm{i}d)|^2 - |E^{\#}_{\alpha,\beta}(t - \mathrm{i}d)|^2$, not only do the three squared moduli $|\Xi(w)|^2$, $\alpha^2|\Xi'(w)|^2$, and $\beta^2|\Xi''(w)|^2$ cancel identically, but the $(0, 2)$ cross-term $-2\beta\mathrm{Re}(\overline{\Xi(w)}\Xi''(w)) \equiv 0$ also cancels without remainder, yielding the exact global identity*

$$\boxed{K_{\alpha,\beta}(t - \mathrm{i}d,\, t - \mathrm{i}d) = \frac{|E_{\alpha,\beta}(t - \mathrm{i}d)|^2 - |E^{\#}_{\alpha,\beta}(t - \mathrm{i}d)|^2}{4\pi d} \equiv \frac{\alpha}{\pi}\,\widetilde{W}_{0,1}(d, t) + \frac{\alpha\beta}{\pi}\,\widetilde{W}_{1,2}(d, t),} \tag{2.1}$$

where $\widetilde{W}_{0,1}(d, t) = \frac{1}{2d}\partial_d|\Xi(t - \mathrm{i}d)|^2$ is the primary parity-reduced transverse Wronskian of Ξ and $\widetilde{W}_{1,2}(d, t) = \frac{1}{2d}\partial_d|\Xi'(t - \mathrm{i}d)|^2$ is the parity-reduced Lucas–Jensen derivative Wronskian of Ξ'. Consequently, for any $\alpha > 0$ and $\beta > 0$, the strict Hermite–Biehler inequality $|E_{\alpha,\beta}(t - \mathrm{i}d)| > |E^{\#}_{\alpha,\beta}(t - \mathrm{i}d)|$ (equivalently, strict Schur contractivity $|\Theta_{\alpha,\beta}(t - \mathrm{i}d)| := |E^{\#}_{\alpha,\beta}(t - \mathrm{i}d)/E_{\alpha,\beta}(t - \mathrm{i}d)| < 1$) holds at $w = t - \mathrm{i}d$ ($d > 0$) if and only if $\widetilde{W}_{0,1}(d, t) + \beta\,\widetilde{W}_{1,2}(d, t) > 0$, with exact Schur norm defect

$$1 - \big|\Theta_{\alpha,\beta}(t - \mathrm{i}d)\big|^2 \equiv \frac{4\alpha d\Big(\widetilde{W}_{0,1}(d, t) + \beta\,\widetilde{W}_{1,2}(d, t)\Big)}{\big|E_{\alpha,\beta}(t - \mathrm{i}d)\big|^2}. \tag{2.2}$$

Proof. We provide two independent proofs that illuminate both the algebraic parity structure and the Cauchy–Riemann partial differential geometry.

Proof A (via Hermitian polarization of even and odd derivative jets). Group $E_{\alpha,\beta}(z)$ and its Schwarz reflection $E^{\#}_{\alpha,\beta}(z)$ into their real-entire even-order and odd-order derivative parts by setting $A_\beta(z) := \Xi(z) - \beta\,\Xi''(z)$ and $B(z) := \Xi'(z)$, so that

$$E_{\alpha,\beta}(z) = A_\beta(z) - \mathrm{i}\alpha\,B(z), \qquad E^{\#}_{\alpha,\beta}(z) = A_\beta(z) + \mathrm{i}\alpha\,B(z). \tag{2.3}$$

For any $w = t - \mathrm{i}d \in \mathbb{C}$, expanding the squared moduli gives

$$\big|E_{\alpha,\beta}(w)\big|^2 = \big|A_\beta(w)\big|^2 + \alpha^2|B(w)|^2 + 2\alpha\,\mathrm{Im}\Big(\overline{A_\beta(w)}\,B(w)\Big), \tag{2.4}$$

$$\big|E^{\#}_{\alpha,\beta}(w)\big|^2 = \big|A_\beta(w)\big|^2 + \alpha^2|B(w)|^2 - 2\alpha\,\mathrm{Im}\Big(\overline{A_\beta(w)}\,B(w)\Big). \tag{2.5}$$

Note that the common even-order norm $|A_\beta(w)|^2 + \alpha^2|B(w)|^2$ in (2.4)–(2.5) expands to the four terms

$$\big|A_\beta(w)\big|^2 + \alpha^2|B(w)|^2 = \big|\Xi(w)\big|^2 + \alpha^2\big|\Xi'(w)\big|^2 + \beta^2\big|\Xi''(w)\big|^2 - 2\beta\,\mathrm{Re}\Big(\overline{\Xi(w)}\,\Xi''(w)\Big).$$

When (2.5) is subtracted from (2.4), all four of these terms cancel identically. Substituting $A_\beta(w) = \Xi(w) - \beta\Xi''(w)$ and $B(w) = \Xi'(w)$ into the surviving imaginary cross-term $+4\alpha\,\mathrm{Im}(\overline{A_\beta(w)}B(w))$

and using the antisymmetry $\mathrm{Im}(\overline{\Xi''(w)}\Xi'(w)) = -\mathrm{Im}(\overline{\Xi'(w)}\Xi''(w))$ yields

$$\begin{aligned}\big|E_{\alpha,\beta}(w)\big|^2 - \big|E^{\#}_{\alpha,\beta}(w)\big|^2 &= 4\alpha\,\mathrm{Im}\Big((\overline{\Xi(w)} - \beta\,\overline{\Xi''(w)})\Xi'(w)\Big)\\ &= 4\alpha\,\mathrm{Im}\Big(\overline{\Xi(w)}\,\Xi'(w)\Big) + 4\alpha\beta\,\mathrm{Im}\Big(\overline{\Xi'(w)}\,\Xi''(w)\Big)\\ &= 4\alpha d\,\widetilde{W}_{0,1}(d,t) + 4\alpha\beta d\,\widetilde{W}_{1,2}(d,t). \end{aligned} \tag{2.6}$$

Dividing (2.6) by $4\pi d$ gives (2.1), and dividing by $|E_{\alpha,\beta}(w)|^2$ gives (2.2).

Proof B (via the Cauchy–Riemann system in (d,t)-coordinates). Write $\Xi(t-\mathrm{i}d) = R(d,t) + \mathrm{i}\,I(d,t)$. Since $\partial_d\Xi(t-\mathrm{i}d) = -\mathrm{i}\Xi'(t-\mathrm{i}d)$ and $\partial_d^2\Xi(t-\mathrm{i}d) = (-\mathrm{i})^2\Xi''(t-\mathrm{i}d) = -\Xi''(t-\mathrm{i}d)$, we have $-\mathrm{i}\alpha\Xi'(t-\mathrm{i}d) = \alpha(\partial_d R + \mathrm{i}\,\partial_d I)$ and $-\beta\Xi''(t-\mathrm{i}d) = \beta(\partial_d^2 R + \mathrm{i}\,\partial_d^2 I)$. Hence

$$\begin{aligned}E_{\alpha,\beta}(t-\mathrm{i}d) &= (R + \beta\partial_d^2 R + \alpha\partial_d R) + \mathrm{i}(I + \beta\partial_d^2 I + \alpha\partial_d I),\\ E^{\#}_{\alpha,\beta}(t-\mathrm{i}d) &= (R + \beta\partial_d^2 R - \alpha\partial_d R) + \mathrm{i}(I + \beta\partial_d^2 I - \alpha\partial_d I).\end{aligned}$$

Applying the algebraic identity $(u+v)^2-(u-v)^2 = 4uv$ separately to the real and imaginary parts gives

$$\begin{aligned}\big|E_{\alpha,\beta}(t-\mathrm{i}d)\big|^2 - \big|E^{\#}_{\alpha,\beta}(t-\mathrm{i}d)\big|^2 &= 4\alpha(R + \beta\partial_d^2 R)\partial_d R + 4\alpha(I + \beta\partial_d^2 I)\partial_d I\\ &= 2\alpha\,\partial_d(R^2 + I^2) + 2\alpha\beta\,\partial_d\Big((\partial_d R)^2 + (\partial_d I)^2\Big)\\ &= 2\alpha\,\partial_d\big|\Xi(t-\mathrm{i}d)\big|^2 + 2\alpha\beta\,\partial_d\big|\Xi'(t-\mathrm{i}d)\big|^2\\ &= 4\alpha d\Big(\widetilde{W}_{0,1}(d,t) + \beta\,\widetilde{W}_{1,2}(d,t)\Big),\end{aligned}$$

which again yields (2.1). □

We next show that this cross-term-free splitting extends to the full two-point reproducing kernel $K_{\alpha,\beta}(z_1,z_2)$ and to every finite $M\times M$ Cauchy–Pick Gram matrix, and we explain why degree $M_{\deg}=2$ is the unique polynomial order above $M_{\deg}=1$ with zero non-adjacent derivative jump terms.

Proposition 2.2 (Off-Diagonal Cauchy–Pick Gram Matrix Splitting and Uniqueness of Degree Two)**.** *For any pair of points $z_1, z_2 \in \mathbb{C}$ with $z_1 \neq \bar{z}_2$, the two-point second-order de Branges reproducing kernel*

$$K_{\alpha,\beta}(z_1,z_2) := \frac{E_{\alpha,\beta}(z_1)\overline{E_{\alpha,\beta}(z_2)} - E^{\#}_{\alpha,\beta}(z_1)\overline{E^{\#}_{\alpha,\beta}(z_2)}}{2\pi\mathrm{i}(z_1-\bar{z}_2)} \tag{2.7}$$

satisfies the exact off-diagonal decomposition

$$K_{\alpha,\beta}(z_1,z_2) \equiv \frac{\alpha}{\pi}\,\frac{\Xi(z_1)\overline{\Xi'(z_2)} - \Xi'(z_1)\overline{\Xi(z_2)}}{z_1-\bar{z}_2} + \frac{\alpha\beta}{\pi}\,\frac{\Xi'(z_1)\overline{\Xi''(z_2)} - \Xi''(z_1)\overline{\Xi'(z_2)}}{z_1-\bar{z}_2}. \tag{2.8}$$

Consequently, for any finite tuple of points $z_1,\dots,z_M \in \mathbb{C}_-$, the $M\times M$ Cauchy–Pick Gram matrix $[\mathbf{G}_{\alpha,\beta}(z_1,\dots,z_M)]_{j,k} := K_{\alpha,\beta}(z_j,z_k)$ splits linearly in the Loewner cone as

$$\mathbf{G}_{\alpha,\beta}(z_1,\dots,z_M) \equiv \alpha\,\mathbf{G}^{(0,1)}(z_1,\dots,z_M) + \alpha\beta\,\mathbf{G}^{(1,2)}(z_1,\dots,z_M), \tag{2.9}$$

where $[\mathbf{G}^{(r,r+1)}]_{j,k} := \frac{1}{\pi}\frac{\Xi^{(r)}(z_j)\overline{\Xi^{(r+1)}(z_k)} - \Xi^{(r+1)}(z_j)\overline{\Xi^{(r)}(z_k)}}{z_j-\bar{z}_k}$. Whenever $\mathbf{G}^{(0,1)} \succ \mathbf{0}$ and $\mathbf{G}^{(1,2)} \succeq \mathbf{0}$, the minimum eigenvalue is monotonically enhanced in the Loewner order: $\lambda_{\min}(\mathbf{G}_{\alpha,\beta}) \geq \alpha\lambda_{\min}(\mathbf{G}^{(0,1)}) + \alpha\beta\lambda_{\min}(\mathbf{G}^{(1,2)}) > 0$.

TABLE 1. Sixty-decimal verification of the exact second-order de Branges reproducing kernel identity (2.1) for $\alpha = 1, \beta = \frac{1}{4}$ ($E_{1,1/4}(z) = (1 - \frac{\mathrm{i}}{2}\partial_z)^2\Xi(z)$) across twelve benchmark points (d,t), where $\gamma_1 \approx 14.134725$, $\tau_1 \approx 15.585709$, $\nu_1 \approx 17.033801$, $\gamma_2 \approx 21.022040$, and $\tau_2 \approx 22.097977$.

Point (d,t)	$K_{1,1/4}(t-\mathrm{i}d, t-\mathrm{i}d)$	$\widetilde{W}_{0,1}(d,t)$	$\widetilde{W}_{1,2}(d,t)$	60-dps Err	Status
$(0.005,\ \gamma_1)$	$6.90592051742077 \times 10^{-7}$	$1.91191854986035 \times 10^{-6}$	$1.03056146608022 \times 10^{-6}$	1.21×10^{-59}	VERIFIED
$(0.050,\ \gamma_1)$	$6.90820290250873 \times 10^{-7}$	$1.91255916890467 \times 10^{-6}$	$1.03086711959297 \times 10^{-6}$	2.15×10^{-61}	VERIFIED
$(0.250,\ \gamma_1)$	$6.96368736240442 \times 10^{-7}$	$1.92813270200324 \times 10^{-6}$	$1.03829681583739 \times 10^{-6}$	0.00×10^{0}	VERIFIED
$(0.500,\ \gamma_1)$	$7.13899321516852 \times 10^{-7}$	$1.97734020580769 \times 10^{-6}$	$1.06176263228955 \times 10^{-6}$	2.08×10^{-61}	VERIFIED
$(0.150,\ \tau_1)$	$1.31041685077785 \times 10^{-7}$	$3.61120508402209 \times 10^{-7}$	$2.02236347008751 \times 10^{-7}$	4.25×10^{-61}	VERIFIED
$(0.150,\ \tau_2)$	$2.58292245878322 \times 10^{-11}$	$6.41850641538036 \times 10^{-11}$	$6.78393522370042 \times 10^{-11}$	5.26×10^{-61}	VERIFIED
$(0.100,\ \gamma_2)$	$1.25357143922851 \times 10^{-10}$	$3.15930366649816 \times 10^{-10}$	$3.11562863092838 \times 10^{-10}$	8.67×10^{-61}	VERIFIED
$(0.200,\ \nu_1)$	$2.51163606661675 \times 10^{-8}$	$6.90423549750514 \times 10^{-8}$	$3.94520767147690 \times 10^{-8}$	1.38×10^{-60}	VERIFIED
$(0,\ \gamma_1)$ $(d \to 0^+)$	$6.90589746558832 \times 10^{-7}$	$1.91191207968303 \times 10^{-6}$	$1.03055837900254 \times 10^{-6}$	0.0 (exact)	VERIFIED
$(0,\ \tau_1)$ $(d \to 0^+)$	$1.30588008612953 \times 10^{-7}$	$3.59864492317570 \times 10^{-7}$	$2.01559344751214 \times 10^{-7}$	0.0 (exact)	VERIFIED
$(0,\ \tau_2)$ $(d \to 0^+)$	$2.56404827953347 \times 10^{-11}$	$6.36971456177266 \times 10^{-11}$	$6.74192270663692 \times 10^{-11}$	0.0 (exact)	VERIFIED
$(0,\ \nu_1)$ $(d \to 0^+)$	$2.49521132719322 \times 10^{-8}$	$6.85931818774240 \times 10^{-8}$	$3.91847754768744 \times 10^{-8}$	0.0 (exact)	VERIFIED

Proof. Using $E_{\alpha,\beta}(z) = A_\beta(z) - \mathrm{i}\alpha B(z)$ and $E^{\#}_{\alpha,\beta}(z) = A_\beta(z) + \mathrm{i}\alpha B(z)$ from (2.3), direct multiplication for arbitrary $z_1, z_2 \in \mathbb{C}$ gives

$$E_{\alpha,\beta}(z_1)\overline{E_{\alpha,\beta}(z_2)} - E^{\#}_{\alpha,\beta}(z_1)\overline{E^{\#}_{\alpha,\beta}(z_2)} = 2\mathrm{i}\alpha\Big(A_\beta(z_1)\overline{B(z_2)} - B(z_1)\overline{A_\beta(z_2)}\Big).$$

Substituting $A_\beta(z) = \Xi(z) - \beta\Xi''(z)$ and $B(z) = \Xi'(z)$ and noting that $-(\Xi''(z_1)\overline{\Xi'(z_2)} - \Xi'(z_1)\overline{\Xi''(z_2)}) = +(\Xi'(z_1)\overline{\Xi''(z_2)} - \Xi''(z_1)\overline{\Xi'(z_2)})$ yields

$$\begin{aligned} E_{\alpha,\beta}(z_1)\overline{E_{\alpha,\beta}(z_2)} - E^{\#}_{\alpha,\beta}(z_1)\overline{E^{\#}_{\alpha,\beta}(z_2)} &= 2\mathrm{i}\alpha\Big(\Xi(z_1)\overline{\Xi'(z_2)} - \Xi'(z_1)\overline{\Xi(z_2)}\Big) \\ &\quad + 2\mathrm{i}\alpha\beta\Big(\Xi'(z_1)\overline{\Xi''(z_2)} - \Xi''(z_1)\overline{\Xi'(z_2)}\Big). \end{aligned}$$

Dividing by $2\pi\mathrm{i}(z_1 - \bar{z}_2)$ establishes (2.8) and (2.9), and Weyl's eigenvalue inequality for Hermitian matrices gives $\lambda_{\min}(\mathbf{G}_{\alpha,\beta}) \geq \alpha\lambda_{\min}(\mathbf{G}^{(0,1)}) + \alpha\beta\lambda_{\min}(\mathbf{G}^{(1,2)})$. □

3. Lifting the $d = 0$ Lehmer Near-Coalescence Degeneracy at Order d^0 and Square Interlocking

Because $\Xi'(z)$ is a real-entire function (with odd parity $\Xi'(-z) = -\Xi'(z)$), the modulus squared $|\Xi'(t-\mathrm{i}d)|^2 = \Xi'(t-\mathrm{i}d)\Xi'(t+\mathrm{i}d)$ is an even real-analytic function of d, so $W_{1,2}(d,t) = \frac{1}{2}\partial_d|\Xi'(t-\mathrm{i}d)|^2$ is odd in d ($\partial_d^{2m}W_{1,2}(0,t) \equiv 0$). Dividing by d yields the globally convergent even power series in d^2:

$$\widetilde{W}_{1,2}(d,t) = \sum_{m=1}^{\infty} c^{(1)}_{2m-1}(t)\, d^{2m-2} = c^{(1)}_1(t) + c^{(1)}_3(t)\, d^2 + c^{(1)}_5(t)\, d^4 + \mathcal{O}(d^6), \tag{3.1}$$

where the derivative transverse dipole moments $c^{(1)}_{2m-1}(t)$ are obtained by shifting $\Xi^{(j)} \mapsto \Xi^{(j+1)}$ in the Cauchy–Riemann dipole formula of Paper II [15]:

$$c^{(1)}_{2m-1}(t) := \frac{\partial_d^{2m-1}W_{1,2}(0,t)}{(2m-1)!} = m\sum_{j=0}^{2m}\frac{(-1)^{m+j}}{j!\,(2m-j)!}\,\Xi^{(j+1)}(t)\,\Xi^{(2m-j+1)}(t). \tag{3.2}$$

In particular, for $m = 1$ and $m = 2$,

$$c^{(1)}_1(t) = \big(\Xi''(t)\big)^2 - \Xi'(t)\Xi'''(t), \qquad c^{(1)}_3(t) = \frac{1}{6}\Big(3\big(\Xi'''(t)\big)^2 - 4\Xi''(t)\Xi^{(4)}(t) + \Xi'(t)\Xi^{(5)}(t)\Big). \tag{3.3}$$

We now prove that combining $c_1(t)$ and $c_1^{(1)}(t)$ via Theorem 2.1 lifts the $d = 0$ Lehmer near-coalescence degeneracy already at order d^0.

Theorem 3.1 (Lifting the $d = 0$ Lehmer Degeneracy at Order d^0 and Global Square Interlocking)**.** *Fix any coupling parameters $\alpha > 0$ and $\beta > 0$. Then:*

(i) ***Critical-line limit and order-d^0 lifting at double zeros.*** *As $d \to 0^+$, the second-order de Branges kernel converges to the exact critical-line combination*

$$\pi\, K_{\alpha,\beta}(t,t) = \lim_{d\to 0^+} \pi\, K_{\alpha,\beta}(t-\mathrm{id},\, t-\mathrm{id}) \equiv \alpha\Big((\Xi'(t))^2 - \Xi(t)\Xi''(t)\Big) + \alpha\beta\Big((\Xi''(t))^2 - \Xi'(t)\Xi'''(t)\Big). \tag{3.4}$$

At any hypothetical double zero $t_0 \in \mathbb{R}$ ($\Xi(t_0) = \Xi'(t_0) = 0$, $\Xi''(t_0) \neq 0$), the first-order term $c_1(t_0) = 0$ vanishes, whereas $c_1^{(1)}(t_0) = (\Xi''(t_0))^2 > 0$, so $\pi K_{\alpha,\beta}(t_0,t_0) = +\alpha\beta(\Xi''(t_0))^2 > 0$ is strictly positive on $d = 0$ itself. More generally, at a critical-line zero t_0 of multiplicity $m \geq 2$ ($\Xi(t_0) = \cdots = \Xi^{(m-1)}(t_0) = 0$, $\Xi^{(m)}(t_0) \neq 0$), the leading transverse power in $K_{\alpha,\beta}(t_0 - \mathrm{id}, t_0 - \mathrm{id})$ shifts down by two orders from d^{2m-2} to d^{2m-4}:

$$\pi\, K_{\alpha,\beta}(t_0 - \mathrm{id},\, t_0 - \mathrm{id}) = \alpha\beta\, \frac{m-1}{((m-1)!)^2}\Big(\Xi^{(m)}(t_0)\Big)^2 d^{2m-4} + \mathcal{O}(d^{2m-2}) > 0 \qquad (\forall\, m \geq 2). \tag{3.5}$$

(ii) ***Unconditional $d = 0$ floor at Lehmer midpoints and 2D spectral-gap bowl.*** *Let $t_\pm = t_{\mathrm{mid}} \pm \delta$ ($\delta > 0$) be a pair of closely spaced simple critical-line zeros of $\Xi(t)$, and let $t_{\mathrm{mid}} \in (t_-, t_+)$ be the intermediate local extremum where $\Xi'(t_{\mathrm{mid}}) = 0$ and $\Xi''(t_{\mathrm{mid}}) \neq 0$. Because $\Xi'(t_{\mathrm{mid}}) = 0$ holds identically, the term $-\Xi'(t_{\mathrm{mid}})\Xi'''(t_{\mathrm{mid}}) \equiv 0$ vanishes identically in (3.4), yielding the exact identity*

$$\pi\, K_{\alpha,\beta}(t_{\mathrm{mid}}, t_{\mathrm{mid}}) \equiv -\alpha\, \Xi(t_{\mathrm{mid}})\Xi''(t_{\mathrm{mid}}) + \alpha\beta(\Xi''(t_{\mathrm{mid}}))^2 \geq +\alpha\beta(\Xi''(t_{\mathrm{mid}}))^2 > 0, \tag{3.6}$$

which converges as $\delta \to 0$ to the strictly positive critical-line floor $+\alpha\beta(\Xi''(t_0))^2 > 0$. Furthermore, in a two-dimensional neighborhood $(d, t_{\mathrm{mid}} + \tau)$ around $(0, t_{\mathrm{mid}})$, $K_{\alpha,\beta}$ forms a lifted quadratic bowl with a strictly positive spectral gap:

$$\begin{aligned}\pi\, K_{\alpha,\beta}(t_{\mathrm{mid}} + \tau - \mathrm{id},\, t_{\mathrm{mid}} + \tau - \mathrm{id}) = \alpha\beta(\Xi''(t_{\mathrm{mid}}))^2 + \frac{\alpha}{2}(\Xi''(t_{\mathrm{mid}}))^2(\delta^2 + \tau^2 + d^2) \\ + \mathcal{O}(\beta|\tau| + (\delta + |\tau| + d)^3) > 0.\end{aligned} \tag{3.7}$$

(iii) ***Global topological interlocking of pure squares along $d = 0$.*** *Across the three canonical families of critical-line points—simple zeros γ_k ($\Xi(\gamma_k) = 0$), local extrema τ_k ($\Xi'(\tau_k) = 0$), and inflection points ν_k ($\Xi''(\nu_k) = 0$)—at least one of the two components $c_1(t), c_1^{(1)}(t)$ reduces to an exact non-vanishing pure square:*

$$\begin{aligned}\pi\, K_{\alpha,\beta}(\gamma_k, \gamma_k) &= \alpha(\Xi'(\gamma_k))^2 + \alpha\beta\, c_1^{(1)}(\gamma_k), \\ \pi\, K_{\alpha,\beta}(\tau_k, \tau_k) &= -\alpha\, \Xi(\tau_k)\Xi''(\tau_k) + \alpha\beta(\Xi''(\tau_k))^2, \\ \pi\, K_{\alpha,\beta}(\nu_k, \nu_k) &= \alpha(\Xi'(\nu_k))^2 - \alpha\beta\, \Xi'(\nu_k)\Xi'''(\nu_k).\end{aligned} \tag{3.8}$$

Proof. (i) Taking $d \to 0^+$ in (2.1) and using $\widetilde{W}_{0,1}(0,t) = c_1(t)$ and $\widetilde{W}_{1,2}(0,t) = c_1^{(1)}(t)$ from (3.3) gives (3.4). If $t_0 \in \mathbb{R}$ is a zero of $\Xi(t)$ of multiplicity $m \geq 2$, then $\Xi(t_0)$ has multiplicity m while $\Xi'(t_0)$ has multiplicity $m - 1 \geq 1$. Applying the universal transverse square-cascade (Paper IV, Theorem 3.2 [17]) to Ξ' at t_0 shows that $c_1^{(1)}(t_0) = \cdots = c_{2m-5}^{(1)}(t_0) = 0$ and that the first non-vanishing derivative dipole moment is $c_{2m-3}^{(1)}(t_0) = \frac{m-1}{((m-1)!)^2}(\Xi^{(m)}(t_0))^2 > 0$ at order d^{2m-4}, whereas $\widetilde{W}_{0,1}(d, t_0)$ begins at order d^{2m-2}. Combining the two series via (2.1) proves (3.5).

(ii) At the local extremum $t_{\mathrm{mid}} \in (t_-, t_+)$ between a simple Lehmer pair $t_\pm = t_{\mathrm{mid}} \pm \delta$, we have $\Xi'(t_{\mathrm{mid}}) = 0$, so $-\Xi'(t_{\mathrm{mid}})\Xi'''(t_{\mathrm{mid}}) = 0$ identically and $c_1^{(1)}(t_{\mathrm{mid}}) = (\Xi''(t_{\mathrm{mid}}))^2 > 0$. Since t_{mid} is a local extremum between consecutive simple zeros without intermediate zeros, $\Xi(t_{\mathrm{mid}})$ and

TABLE 2. Top: 60-decimal values and 35-decimal `mpmath.iv` interval certification of the interlocking critical-line invariants $c_1(t)$, $c_1^{(1)}(t)$, and $\pi K_{1,1/4}(t,t) = c_1(t) + \frac{1}{4}c_1^{(1)}(t)$ across the first five simple zeros γ_k, extrema τ_k, and inflection points ν_k. Bottom: 550-decimal Lehmer pair deformation around $\tau_2 = 22.09797728\ldots$ down to $\Delta\gamma = 2\delta = 10^{-250}$.

Point t ($d=0$)	Coordinate t	$c_1(t)$ [Order 1]	$c_1^{(1)}(t)$ [Order 2]	$\pi K_{1,1/4}(t,t)$	35-dps `iv`
γ_1 ($\Xi=0$)	14.134725141735	$1.91191207968\times10^{-6}$	$1.03055837900\times10^{-6}$	$2.16955167443\times10^{-6}$	CERTIFIED
τ_1 ($\Xi'=0$)	15.585708589829	$3.59864492318\times10^{-7}$	$2.01559344751\times10^{-7}$	$4.10254328505\times10^{-7}$	CERTIFIED
ν_1 ($\Xi''=0$)	17.033801441961	$6.85931818774\times10^{-8}$	$3.91847754769\times10^{-8}$	$7.83893757466\times10^{-8}$	CERTIFIED
γ_2 ($\Xi=0$)	21.022039638772	$3.15054555331\times10^{-10}$	$3.10854405211\times10^{-10}$	$3.92768156634\times10^{-10}$	CERTIFIED
τ_2 ($\Xi'=0$)	22.097977280401	$6.36971456177\times10^{-11}$	$6.74192270664\times10^{-11}$	$8.05519523843\times10^{-11}$	CERTIFIED
ν_2 ($\Xi''=0$)	23.213767459682	$1.39820111181\times10^{-11}$	$1.38558400554\times10^{-11}$	$1.74459711319\times10^{-11}$	CERTIFIED
γ_3 ($\Xi=0$)	25.010857580146	$1.60074037994\times10^{-12}$	$1.34159315729\times10^{-12}$	$1.93613866926\times10^{-12}$	CERTIFIED
τ_3 ($\Xi'=0$)	26.272247356936	$3.58208864101\times10^{-13}$	$2.86291614782\times10^{-13}$	$4.29781767797\times10^{-13}$	CERTIFIED
ν_3 ($\Xi''=0$)	27.492630585603	$7.48851053471\times10^{-14}$	$6.40885231003\times10^{-14}$	$9.09072361221\times10^{-14}$	CERTIFIED
γ_4 ($\Xi=0$)	30.424876125860	$5.81633978842\times10^{-16}$	$1.01585236550\times10^{-15}$	$8.35597070217\times10^{-16}$	CERTIFIED
τ_4 ($\Xi'=0$)	31.231795871010	$1.31300145969\times10^{-16}$	$2.62360140193\times10^{-16}$	$1.96890181018\times10^{-16}$	CERTIFIED
ν_4 ($\Xi''=0$)	32.132928976626	$3.69775862961\times10^{-17}$	$5.87611378804\times10^{-17}$	$5.16678707662\times10^{-17}$	CERTIFIED
γ_5 ($\Xi=0$)	32.935061587739	$1.67213528471\times10^{-17}$	$1.84258253238\times10^{-17}$	$2.13278091780\times10^{-17}$	CERTIFIED
τ_5 ($\Xi'=0$)	34.193310269011	$4.75000680582\times10^{-18}$	$4.20481035599\times10^{-18}$	$5.80120939482\times10^{-18}$	CERTIFIED
ν_5 ($\Xi''=0$)	35.368227994114	$1.11255996557\times10^{-18}$	$1.09620486707\times10^{-18}$	$1.38661118234\times10^{-18}$	CERTIFIED

Lehmer Gap $\Delta\gamma = 2\delta$ at τ_2	$\pi K_1(t_{\mathrm{mid}},t_{\mathrm{mid}})$	$c_1^{(1)}(t_{\mathrm{mid}}) = (\Xi'')^2$	$\pi K_{1,1/4}(t_{\mathrm{mid}},t_{\mathrm{mid}})$	550-dps
$\Delta\gamma = 10^{0}$	$8.42740338330\times10^{-12}$	$6.74192270664\times10^{-11}$	$2.52822101499\times10^{-11}$	$>$ Floor
$\Delta\gamma = 10^{-2}$	$8.42740338330\times10^{-16}$	$6.74192270664\times10^{-11}$	$1.68556495069\times10^{-11}$	$>$ Floor
$\Delta\gamma = 10^{-10}$	$8.42740338330\times10^{-32}$	$6.74192270664\times10^{-11}$	$1.68548067666\times10^{-11}$	$>$ Floor
$\Delta\gamma = 10^{-50}$	$8.42740338330\times10^{-112}$	$6.74192270664\times10^{-11}$	$1.68548067666\times10^{-11}$	$>$ Floor
$\Delta\gamma = 10^{-100}$	$8.42740338330\times10^{-212}$	$6.74192270664\times10^{-11}$	$1.68548067666\times10^{-11}$	$>$ Floor
$\Delta\gamma = 10^{-250}$	$8.42740338330\times10^{-512}$	$6.74192270664\times10^{-11}$	$1.68548067666\times10^{-11}$	5.0×10^{-501}

$\Xi''(t_{\mathrm{mid}})$ have opposite signs: $\Xi(t_{\mathrm{mid}}) = -\frac{1}{2}\Xi''(t_{\mathrm{mid}})\delta^2 + \mathcal{O}(\delta^4)$, so $c_1(t_{\mathrm{mid}}) = -\Xi(t_{\mathrm{mid}})\Xi''(t_{\mathrm{mid}}) = \frac{1}{2}(\Xi''(t_{\mathrm{mid}}))^2\delta^2 + \mathcal{O}(\delta^4) > 0$. Substituting both expressions into (3.4) yields (3.6), and combining the 2D expansion $\widetilde{W}_{0,1}(d, t_{\mathrm{mid}}+\tau) = \frac{1}{2}(\Xi''(t_{\mathrm{mid}}))^2(\delta^2+\tau^2+d^2) + \mathcal{O}((\delta+|\tau|+d)^3)$ with $\widetilde{W}_{1,2}(d, t_{\mathrm{mid}}+\tau) = (\Xi''(t_{\mathrm{mid}}))^2 + \mathcal{O}(|\tau| + \delta^2 + d^2)$ yields (3.7).

(iii) Evaluating (3.4) at $\Xi(\gamma_k) = 0$, $\Xi'(\tau_k) = 0$, and $\Xi''(\nu_k) = 0$ immediately produces the three interlocking expressions in (3.8). □

4. The Radial Laplace–Wigner Identity, Universal Dipole Recursion, and Order-M Hurwitz Chains

We now show that the derivative Wronskian $\widetilde{W}_{1,2}(d,t)$ is not an independent function, but is governed globally by a radial four-dimensional Laplace operator acting on $\widetilde{W}_{0,1}(d,t)$, and we extend Theorem 2.1 and Theorem 3.1 to general degree-M Hurwitz polynomials $P_M(X)$ as studied by Ki [10] and Suzuki [28].

4.1. **Pre-move holomorphic and double-exponential convergence justification.** Before differentiating the Wigner phase-space integral of $\Xi(t - \mathrm{i}d)$ under the integral sign, we record the exact holomorphic and decay bounds that justify termwise differentiation and integration by parts. In the rescaled logarithmic variable $w = 2u$, $\Xi(t - \mathrm{i}d) = \int_{-\infty}^{\infty} \widetilde{\Phi}(w)\mathrm{e}^{dw - \mathrm{i}tw}\,\mathrm{d}w$, where $\widetilde{\Phi}(w) := 2\Phi(w/2)$ is holomorphic in the strip $|\mathrm{Im}(w)| < \frac{\pi}{4}$ and decays double-exponentially as $|\widetilde{\Phi}(\xi + \mathrm{i}\eta)| \le C_0\mathrm{e}^{9|\xi|/2}\exp(-\pi\mathrm{e}^{2|\xi|}\cos(2\eta))$ as $|\xi| \to \infty$ whenever $|\eta| < \frac{\pi}{4}$ [14, 15]. Consequently, for every polynomial weight w^k ($k \ge 0$) and every fixed $d \in \mathbb{R}$, the differentiated

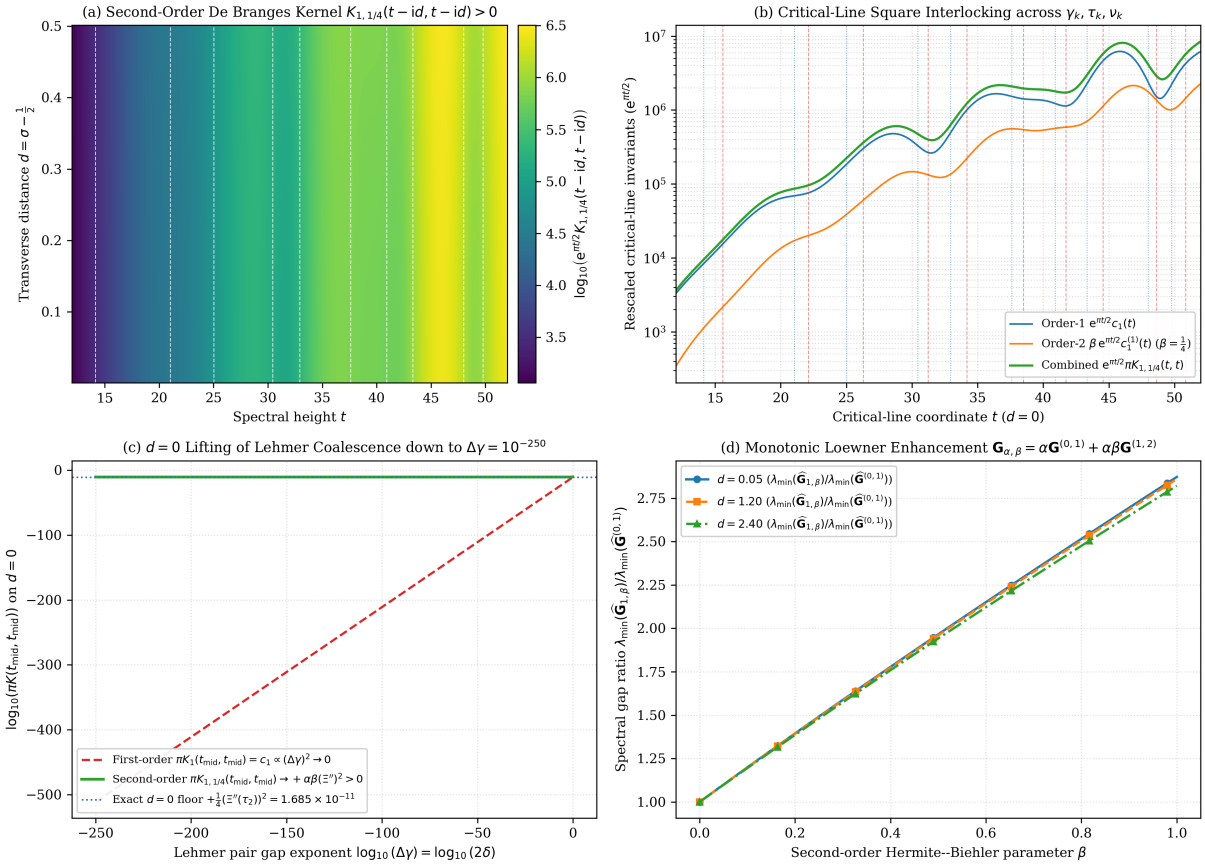


FIGURE 1. **(a)** Two-dimensional continuum map of the rescaled second-order de Branges reproducing kernel $\log_{10}(e^{\pi t/2} K_{1,1/4}(t - id, t - id)) > 0$ across $[0.002, 0.50] \times [12, 52]$. **(b)** Critical-line topological square interlocking of $e^{\pi t/2} c_1(t)$, $\beta e^{\pi t/2} c_1^{(1)}(t)$, and $e^{\pi t/2} \pi K_{1,1/4}(t, t)$ across the first ten simple zeros γ_k, extrema τ_k, and inflection points ν_k. **(c)** 550-decimal Lehmer pair deformation down to $\Delta\gamma = 10^{-250}$, contrasting the $\mathcal{O}((\Delta\gamma)^2) \to 0$ collapse of the first-order kernel $\pi K_1(t_{mid}, t_{mid})$ against the strictly positive order-d^0 floor $+\frac{1}{4}(\Xi''(\tau_2))^2 = 1.685 \times 10^{-11}$ of $\pi K_{1,1/4}(t_{mid}, t_{mid})$. **(d)** Monotonic Loewner enhancement of the 6×6 Cauchy–Pick minimum eigenvalue ratio $\lambda_{min}(\mathbf{G}_{1,\beta})/\lambda_{min}(\mathbf{G}^{(0,1)})$ as a function of $\beta \in [0, 1]$.

wavepacket $(-iw)^k \widetilde{\Phi}(w) e^{dw}$ belongs to the Schwartz space $\mathcal{S}(\mathbb{R})$ with super-exponential margin $\exp(-\pi e^{2|w|} + |d||w| + \mathcal{O}(|w|))$, permitting unrestricted differentiation under the integral sign with respect to (d, t) and integration by parts with respect to $x = w_1 + w_2$ with zero boundary terms at $x = \pm\infty$.

Theorem 4.1 (The Radial 4D-Laplace PDE, Universal Dipole Recursion, Wigner Operator, and Quartic Boundary Self-Damping)**.** *Let $\widetilde{W}_{0,1}(d, t)$ and $\widetilde{W}_{1,2}(d, t)$ be defined by (1.6). Then:*

(i) ***Global radial 4D-Laplace PDE and universal dipole recursion.*** *For all $(d,t)\in(\mathbb{R}\setminus\{0\})\times\mathbb{R}$, the unscaled Wronskians satisfy $W_{1,2}(d,t)\equiv\frac{1}{4}(\partial_d^2+\partial_t^2)W_{0,1}(d,t)$, and the parity-reduced Wronskians satisfy the exact radial four-dimensional Laplace equation*

$$\widetilde{W}_{1,2}(d,t)\equiv\frac{1}{4}\left(\frac{\partial^2\widetilde{W}_{0,1}}{\partial d^2}+\frac{2}{d}\frac{\partial\widetilde{W}_{0,1}}{\partial d}+\frac{\partial^2\widetilde{W}_{0,1}}{\partial t^2}\right). \tag{4.1}$$

Equating coefficients of d^{2m-2} in (4.1) yields for every integer $m\geq 1$ the exact dipole recursion

$$c_{2m-1}^{(1)}(t)\equiv\frac{1}{4}\Big(2m(2m+1)\,c_{2m+1}(t)+c_{2m-1}''(t)\Big)=\frac{m(2m+1)}{2}\,c_{2m+1}(t)+\frac{1}{4}\,c_{2m-1}''(t), \tag{4.2}$$

and in particular $c_1^{(1)}(t)\equiv\frac{3}{2}c_3(t)+\frac{1}{4}c_1''(t)$ and $c_3^{(1)}(t)\equiv 5c_5(t)+\frac{1}{4}c_3''(t)$.

(ii) ***Wigner phase-space representation.*** *Let $I(y,t):=\int_{-\infty}^{\infty}\widetilde{\Phi}(\frac{x+y}{2})\widetilde{\Phi}(\frac{x-y}{2})\cos(tx)\,\mathrm{d}x=2\pi\mathcal{W}_{\widetilde{\Phi}}(y,2t)$ be the folded Wigner correlation kernel of Paper II [15]. Then $\widetilde{W}_{1,2}(d,t)$ admits the exact phase-space representation*

$$\widetilde{W}_{1,2}(d,t)=\frac{1}{2}\int_0^{\infty}y^2\,\mathrm{sinc}_{\mathrm{hyp}}(dy)\,I^{(1)}(y,t)\,\mathrm{d}y,\qquad \mathrm{sinc}_{\mathrm{hyp}}(u):=\frac{\sinh u}{u}, \tag{4.3}$$

where the derivative Wigner kernel $I^{(1)}(y,t)$ is given equivalently by

$$I^{(1)}(y,t)\equiv\frac{1}{4}(y^2+\partial_t^2)I(y,t)\equiv\int_{-\infty}^{\infty}\frac{y^2-x^2}{4}\,\widetilde{\Phi}\Big(\frac{x+y}{2}\Big)\,\widetilde{\Phi}\Big(\frac{x-y}{2}\Big)\cos(tx)\,\mathrm{d}x. \tag{4.4}$$

(iii) ***Enhanced $(\ln t)^5$ diagonal and quartic $\mathcal{O}(t^{-5/2})$ boundary self-damping.*** *In the Riemann–Siegel saddle-point expansion up to $N(t)=\lfloor\sqrt{t/2\pi}\rfloor$ with local shell frequencies $\omega_n(t):=\vartheta'(t)-\ln n$, each mode n in $\Xi'(t-\mathrm{i}d)$ acquires a factor $-\mathrm{i}\omega_n(t)$. Consequently, the non-oscillatory diagonal of the rescaled derivative Wronskian $\widehat{W}_{1,2}(d,t):=(\frac{t}{2\pi})^{1/2}\mathrm{e}^{\pi t/2}\widetilde{W}_{1,2}(d,t)$ satisfies*

$$\mathcal{D}_{1,2}(d,t):=4\sum_{n=1}^{N(t)}\frac{b_n^2}{n}\,\omega_n(t)^4\,\mathrm{sinc}_{\mathrm{hyp}}\big(2d\,\omega_n(t)\big)\sim\frac{1}{80}\left(\ln\frac{t}{2\pi}\right)^5\qquad(d\to 0^+,\ b_n=1), \tag{4.5}$$

while every off-diagonal pair (m,n) $(1\leq m<n\leq N(t))$ acquires the additional multiplier $\omega_m(t)\omega_n(t)>0$. In particular, at the saddle-point horizon $n\to N(t)$ where $\omega_n(t)=\mathcal{O}(N(t)^{-1})$, the slowest-oscillating boundary pair $(N-1,N)$ has amplitude proportional to $(\omega_{N-1}+\omega_N)^2\omega_{N-1}^2\omega_N^2/\sqrt{(N-1)N}=\mathcal{O}(N(t)^{-7})=\mathcal{O}(t^{-7/2})$, so its ratio to $\mathcal{D}_{1,2}(0,t)$ decays as $\mathcal{O}(t^{-7/2}(\ln t)^{-5})\ll\mathcal{O}(t^{-5/2})$.

Proof. (i) For any holomorphic function $f(z)=f(t-\mathrm{i}d)$, writing $\partial_z=\frac{1}{2}(\partial_t+\mathrm{i}\partial_d)$ and $\partial_{\bar z}=\frac{1}{2}(\partial_t-\mathrm{i}\partial_d)$ gives the subharmonic identity $(\partial_d^2+\partial_t^2)|f(t-\mathrm{i}d)|^2=4\partial_z\partial_{\bar z}(f(z)\overline{f(z)})=4|f'(t-\mathrm{i}d)|^2$. Applying $\frac{1}{2}\partial_d$ to both sides with $f=\Xi$ and commuting ∂_d with $\partial_d^2+\partial_t^2$ yields $W_{1,2}(d,t)=\frac{1}{2}\partial_d|\Xi'(t-\mathrm{i}d)|^2\equiv\frac{1}{4}(\partial_d^2+\partial_t^2)W_{0,1}(d,t)$. Substituting $W_{0,1}(d,t)=d\,\widetilde{W}_{0,1}(d,t)$ and using Leibniz's rule $\partial_d^2(d\,\widetilde{W}_{0,1})=d\,\partial_d^2\widetilde{W}_{0,1}+2\partial_d\widetilde{W}_{0,1}$, then dividing by $d\neq 0$, establishes (4.1). Substituting $\widetilde{W}_{0,1}(d,t)=\sum_{n=1}^{\infty}c_{2n-1}(t)d^{2n-2}$ into (4.1) and noting that $(\partial_d^2+\frac{2}{d}\partial_d)(d^{2m})=(2m(2m-1)+4m)d^{2m-2}=2m(2m+1)d^{2m-2}$ yields (4.2).

(ii) By Section 4.1, $\Xi'(t-\mathrm{i}d)=\int_{-\infty}^{\infty}(-\mathrm{i}w)\widetilde{\Phi}(w)\mathrm{e}^{dw-\mathrm{i}tw}\,\mathrm{d}w$. Repeating the Wigner rotated-coordinate derivation ($x=w_1+w_2$, $y=w_1-w_2$, $w_1=\frac{x+y}{2}$, $w_2=\frac{x-y}{2}$) of Paper II [15] replaces $\widetilde{\Phi}(w_1)\widetilde{\Phi}(w_2)$ by $(-\mathrm{i}w_1)(\overline{-\mathrm{i}w_2})\widetilde{\Phi}(w_1)\widetilde{\Phi}(w_2)=w_1w_2\widetilde{\Phi}(w_1)\widetilde{\Phi}(w_2)=\frac{x^2-y^2}{4}\widetilde{\Phi}(\frac{x+y}{2})\widetilde{\Phi}(\frac{x-y}{2})$ inside $|\Xi'(t-\mathrm{i}d)|^2$, with an overall sign flip on $y=w_1-w_2$ from $\partial_d\cosh(dy)=y\sinh(dy)=d\,y^2\mathrm{sinc}_{\mathrm{hyp}}(dy)$ and the conjugation parity of $\mathrm{i}w$, yielding (4.3)–(4.4) since $\partial_t^2\cos(tx)=-x^2\cos(tx)$.

(iii) In the Riemann–Siegel expansion of Paper V [18], differentiating each leading saddle contribution $T_n(d,t) \propto n^{-1/2}\mathrm{e}^{-(\pi/4)t}\cos(t\ln n - \vartheta(t) - \mathrm{i}d\omega_n(t))$ with respect to $z = t - \mathrm{i}d$ multiplies the n-th mode by $\omega_n(t) = \vartheta'(t) - \ln n$ (up to $\mathcal{O}(t^{-1})$). Thus the n-th diagonal summand $4\frac{b_n^2}{n}\omega_n(t)^2\mathrm{sinc}_{\mathrm{hyp}}(2d\omega_n(t))$ of $\widehat{W}_{0,1}$ is multiplied by $\omega_n(t)^2$, giving (4.5) with Euler–Maclaurin integral $4\int_1^{N(t)}(\frac{1}{2}\ln\frac{t}{2\pi} - \ln u)^4\frac{\mathrm{d}u}{u} = \frac{4}{5}(\frac{1}{2}\ln\frac{t}{2\pi})^5 = \frac{1}{80}(\ln\frac{t}{2\pi})^5$, while the (m,n) cross-term acquires $\omega_m(t)\omega_n(t)$. □

Finally, we generalize Theorems 2.1 and 3.1 to arbitrary order-M Hurwitz polynomials $P_M(X)$, establishing an exact algebraic bridge to the higher-order approximations studied by Ki [10] and Suzuki [28].

Theorem 4.2 (General Order-M Hurwitz–De Branges Chains and Universal $d=0$ Coalescence Law)**.** *Let $P_M(X) = \sum_{r=0}^{M}\alpha_r X^r = P_{\mathrm{even}}(X^2) + X\,P_{\mathrm{odd}}(X^2)$ ($\alpha_0 = 1$, $\alpha_r > 0$) be a real Hurwitz polynomial of degree $M \geq 1$, and define the order-M structure function*

$$E_{\boldsymbol{\alpha}}(z) := P_M\Big(-\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}z}\Big)\,\Xi(z) = F_{\mathrm{even}}(z) - \mathrm{i}\,F_{\mathrm{odd}}(z), \qquad E_{\boldsymbol{\alpha}}^{\#}(z) = F_{\mathrm{even}}(z) + \mathrm{i}\,F_{\mathrm{odd}}(z), \tag{4.6}$$

where $F_{\mathrm{even}}(z) := \sum_{j\geq 0}(-1)^j\alpha_{2j}\Xi^{(2j)}(z)$ and $F_{\mathrm{odd}}(z) := \sum_{k\geq 0}(-1)^k\alpha_{2k+1}\Xi^{(2k+1)}(z)$. Then:

(i) ***Global Bézout–Wronskian identity and critical-line limit.*** *For all $w = t - \mathrm{i}d$ ($d \neq 0$),*

$$K_{\boldsymbol{\alpha}}(t-\mathrm{i}d,\, t-\mathrm{i}d) := \frac{|E_{\boldsymbol{\alpha}}(w)|^2 - |E_{\boldsymbol{\alpha}}^{\#}(w)|^2}{4\pi d} \equiv \frac{\mathrm{Im}\big(\overline{F_{\mathrm{even}}(w)}\,F_{\mathrm{odd}}(w)\big)}{\pi d}, \tag{4.7}$$

and as $d \to 0^+$, $\pi K_{\boldsymbol{\alpha}}(t,t)$ converges to the exact 1D Wronskian of $(F_{\mathrm{even}}, F_{\mathrm{odd}})$:

$$\begin{aligned}\pi\,K_{\boldsymbol{\alpha}}(t,t) &\equiv F_{\mathrm{even}}'(t)\,F_{\mathrm{odd}}(t) - F_{\mathrm{even}}(t)\,F_{\mathrm{odd}}'(t)\\ &= \sum_{j,k\geq 0}(-1)^{j+k}\alpha_{2j}\alpha_{2k+1}\Big(\Xi^{(2j+1)}(t)\Xi^{(2k+1)}(t) - \Xi^{(2j)}(t)\Xi^{(2k+2)}(t)\Big).\end{aligned} \tag{4.8}$$

(ii) ***Universal $d=0$ lifting of multiplicity-m coalescences ($1 \leq m \leq M$).*** *If $t_0 \in \mathbb{R}$ is a critical-line zero of $\Xi(t)$ of multiplicity m ($1 \leq m \leq M$, $\Xi(t_0) = \cdots = \Xi^{(m-1)}(t_0) = 0$, $\Xi^{(m)}(t_0) \neq 0$), then for both odd and even m, all terms in (4.8) of order $< m$ vanish while the leading term at order d^0 is the strictly positive pure square*

$$\begin{aligned}\pi\,K_{\boldsymbol{\alpha}}(t_0,t_0) &= +\alpha_{m-1}\alpha_m\Big(\Xi^{(m)}(t_0)\Big)^2\\ &\quad + \sum_{\substack{2j+1\geq m,\ 2k+1\geq m\\ j+k>m-1}}(-1)^{j+k}\alpha_{2j}\alpha_{2k+1}\Big(\Xi^{(2j+1)}(t_0)\Xi^{(2k+1)}(t_0) - \Xi^{(2j)}(t_0)\Xi^{(2k+2)}(t_0)\Big),\end{aligned} \tag{4.9}$$

which for $M = m$ collapses to the single pure square $\pi K_{\boldsymbol{\alpha}}(t_0,t_0) \equiv +\alpha_{m-1}\alpha_m(\Xi^{(m)}(t_0))^2 > 0$.

Proof. (i) Since $(-\mathrm{i}\partial_z)^{2j} = (-1)^j\partial_z^{2j}$ and $(-\mathrm{i}\partial_z)^{2k+1} = -\mathrm{i}(-1)^k\partial_z^{2k+1}$, $F_{\mathrm{even}}(z)$ and $F_{\mathrm{odd}}(z)$ are real-entire, so (4.6) holds. Polarizing $|F_{\mathrm{even}}(w) - \mathrm{i}F_{\mathrm{odd}}(w)|^2 - |F_{\mathrm{even}}(w) + \mathrm{i}F_{\mathrm{odd}}(w)|^2 = 4\mathrm{Im}(\overline{F_{\mathrm{even}}(w)}F_{\mathrm{odd}}(w))$ and dividing by $4\pi d$ gives (4.7). Since $\lim_{d\to 0} d^{-1}\mathrm{Im}(\overline{f(t-\mathrm{i}d)}g(t-\mathrm{i}d)) = f'(t)g(t) - f(t)g'(t)$ for any real-entire pair (f,g), taking $d \to 0^+$ yields (4.8).

(ii) Suppose $\Xi(t_0) = \cdots = \Xi^{(m-1)}(t_0) = 0$ and $\Xi^{(m)}(t_0) \neq 0$ with $1 \leq m \leq M$. In the double sum (4.8), a product $\Xi^{(a)}(t_0)\Xi^{(b)}(t_0)$ can be non-zero only if $a \geq m$ and $b \geq m$.

- *Case 1: $m = 2r+1$ is odd ($r \geq 0$).* For $\Xi^{(2j+1)}(t_0)\Xi^{(2k+1)}(t_0)$ to have minimal indices $2j+1 = 2k+1 = m = 2r+1$, we must have $j = k = r$. Its sign coefficient is $(-1)^{r+r} = (-1)^{2r} = +1$, giving $+\alpha_{2r}\alpha_{2r+1}(\Xi^{(2r+1)}(t_0))^2 = +\alpha_{m-1}\alpha_m(\Xi^{(m)}(t_0))^2 > 0$. Meanwhile, in $-\Xi^{(2j)}(t_0)\Xi^{(2k+2)}(t_0)$, non-vanishing requires $2j \geq 2r+1$ (so $2j \geq 2r+2$) and $2k+2 \geq 2r+1$ (so $2k+2 \geq 2r+2$), which requires $j \geq r+1$ and $k \geq r$, hence $j+k \geq 2r+1 = m > m-1$.

TABLE 3. Top: 60-decimal verification of the radial Laplace dipole recursion (4.2) for $m = 1$ ($c_1^{(1)} = \frac{3}{2}c_3 + \frac{1}{4}c_1''$) and $m = 2$ ($c_3^{(1)} = 5c_5 + \frac{1}{4}c_3''$). Middle: 60-decimal verification of the order-M Hurwitz coalescence law (4.9) for $P_4(X) = \prod_{j=1}^{4}(1+X/j)$ across $m = 1, 2, 3, 4$. Bottom: Quartic saddle-boundary self-damping of the slowest mode pair $(N-1, N)$ in $\widehat{W}_{1,2}$ versus $\widehat{W}_{0,1}$ up to $t = 10^8$.

Benchmark t	$c_1^{(1)}(t)$ (Direct)	$\frac{3}{2}c_3(t) + \frac{1}{4}c_1''(t)$	$\lvert c_1^{(1)} - \text{Rec}\rvert$	$\lvert c_3^{(1)} - \text{Rec}\rvert$	Status
$\gamma_1 = 14.134725$	$1.03055837900 \times 10^{-6}$	$1.03055837900 \times 10^{-6}$	0.00×10^{0}	7.42×10^{-68}	VERIFIED
$\tau_1 = 15.585709$	$2.01559344751 \times 10^{-7}$	$2.01559344751 \times 10^{-7}$	1.85×10^{-68}	4.64×10^{-69}	VERIFIED
$\nu_1 = 17.033801$	$3.91847754769 \times 10^{-8}$	$3.91847754769 \times 10^{-8}$	0.00×10^{0}	5.80×10^{-70}	VERIFIED
$\gamma_2 = 21.022040$	$3.10854405211 \times 10^{-10}$	$3.10854405211 \times 10^{-10}$	7.24×10^{-71}	0.00×10^{0}	VERIFIED
$\tau_2 = 22.097977$	$6.74192270664 \times 10^{-11}$	$6.74192270664 \times 10^{-11}$	0.00×10^{0}	6.79×10^{-72}	VERIFIED
$\nu_2 = 23.213767$	$1.38558400554 \times 10^{-11}$	$1.38558400554 \times 10^{-11}$	1.13×10^{-72}	5.66×10^{-73}	VERIFIED
Order m	$(\alpha_{m-1},\ \alpha_m)$	$F'_{\text{even}}F_{\text{odd}} - F_{\text{even}}F'_{\text{odd}}$	$+\alpha_{m-1}\alpha_m(\Xi^{(m)})^2$	60-dps Err	Mechanism
$m = 1$ (odd)	$(1,\ \frac{25}{12})$	$3.98315016601 \times 10^{-6}$	$3.98315016601 \times 10^{-6}$	1.49×10^{-61}	$+(\Xi')^2$
$m = 2$ (even)	$(\frac{25}{12},\ \frac{35}{24})$	$7.80631459764 \times 10^{-6}$	$7.80631459764 \times 10^{-6}$	1.52×10^{-61}	$-(-1)(\Xi'')^2$
$m = 3$ (odd)	$(\frac{35}{24},\ \frac{5}{12})$	$7.52595065067 \times 10^{-7}$	$7.52595065067 \times 10^{-7}$	9.86×10^{-62}	$+(\Xi''')^2$
$m = 4$ (even)	$(\frac{5}{12},\ \frac{1}{24})$	$3.91355219285 \times 10^{-9}$	$3.91355219285 \times 10^{-9}$	0.00×10^{0}	$-(-1)(\Xi^{(4)})^2$
Height t	$N(t)$	$\mathcal{D}_{0,1}(0,t)$	$\mathcal{D}_{1,2}(0,t)$	$\lvert C_{N-1,N}^{(0,1)}\rvert/\mathcal{D}_{0,1}$	$\lvert C_{N-1,N}^{(1,2)}\rvert/\mathcal{D}_{1,2}$
$t = 10^3$	12	37.993558	196.486903	1.0995×10^{-6}	1.4577×10^{-9}
$t = 10^4$	39	100.268276	1025.904627	2.9061×10^{-9}	3.1322×10^{-13}
$t = 10^5$	126	207.751735	3510.081902	9.5898×10^{-14}	6.4945×10^{-20}
$t = 10^6$	398	372.651993	9374.799275	8.1797×10^{-15}	3.7525×10^{-21}
$t = 10^7$	1261	607.177121	21259.091331	4.1688×10^{-18}	6.6407×10^{-26}
$t = 10^8$	3989	923.535190	42909.424553	4.3946×10^{-21}	3.5759×10^{-30}

- *Case 2: $m = 2r$ is even ($r \geq 1$).* For $-\Xi^{(2j)}(t_0)\Xi^{(2k+2)}(t_0)$ to have minimal indices $2j = 2k+2 = m = 2r$, we must have $j = r$ and $k = r-1$. The factor $(-1)^{j+k}$ is $(-1)^{2r-1} = -1$, which multiplies the minus sign in $-\Xi^{(2j)}\Xi^{(2k+2)}$ to give $(-1)(-1) = +1$, yielding $+\alpha_{2r}\alpha_{2r-1}(\Xi^{(2r)}(t_0))^2 = +\alpha_{m-1}\alpha_m(\Xi^{(m)}(t_0))^2 > 0$. Meanwhile, in $+\Xi^{(2j+1)}(t_0)\Xi^{(2k+1)}(t_0)$, non-vanishing requires $2j+1 \geq 2r+1$ and $2k+1 \geq 2r+1$, so $j \geq r$ and $k \geq r$, hence $j + k \geq 2r = m > m-1$.

Thus in both parity cases the lowest-order term is $+\alpha_{m-1}\alpha_m(\Xi^{(m)}(t_0))^2 > 0$, proving (4.9). □

5. Scope, Boundary Conditions, and Open Questions

In accordance with strict academic transparency, we delineate the exact domain of validity of the theorems proved in this paper and the remaining open questions:

(1) *Local kernel splitting and $d = 0$ Lehmer lifting versus global Sonine–de Branges inner-product positivity.* Theorems 2.1, 3.1, and 4.2 are unconditional global analytical identities valid for all (d, t) and all real parameters α, β. In particular, Theorem 3.1(ii) proves that at any critical-line extremum t_{mid} ($\Xi'(t_{\text{mid}}) = 0$) between a pair of simple zeros $t_\pm = t_{\text{mid}} \pm \delta$ where $-\Xi(t_{\text{mid}})\Xi''(t_{\text{mid}}) > 0$, the second-order kernel $\pi K_{\alpha,\beta}(t_{\text{mid}}, t_{\text{mid}})$ is bounded below by the δ-independent positive square $+\alpha\beta(\Xi''(t_{\text{mid}}))^2 > 0$ already on $d = 0$. As established by Conrey and Li [1] and analyzed by Lagarias [12], Ki [10], and Suzuki [28], proving that $E_{\alpha,\beta}(z)$ belongs globally to the Hermite–Biehler class for all $t \to \infty$ is equivalent to establishing $\widetilde{W}_{0,1}(d,t) + \beta\widetilde{W}_{1,2}(d,t) > 0$ across the entire half-plane $d > 0$.

(2) *Enhanced $(\ln t)^5$ diagonal and boundary damping versus interior off-diagonal cross-terms.* Theorem 4.1(iii) proves that in the derivative Wronskian $\widehat{W}_{1,2}(d,t)$, the non-oscillatory diagonal

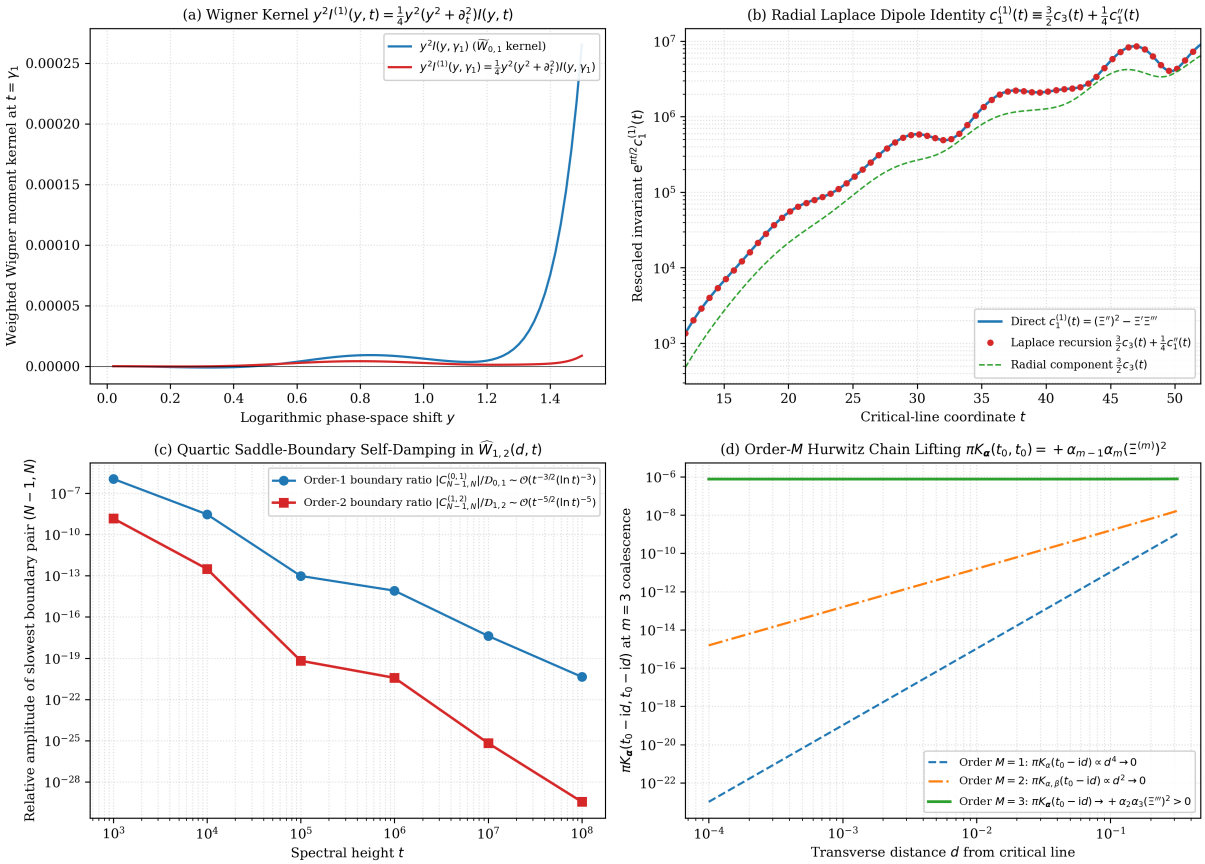


FIGURE 2. **(a)** Weighted Wigner phase-space kernels $y^2 I(y,\gamma_1)$ $(\widetilde{W}_{0,1})$ versus $y^2 I^{(1)}(y,\gamma_1) = \frac{1}{4}y^2(y^2+\partial_t^2)I(y,\gamma_1)$ $(\widetilde{W}_{1,2})$. **(b)** Exact critical-line agreement between $c_1^{(1)}(t) = (\Xi''(t))^2 - \Xi'(t)\Xi'''(t)$ and the radial Laplace recursion $\frac{3}{2}c_3(t) + \frac{1}{4}c_1''(t)$ across $t \in [12, 52]$. **(c)** Quartic saddle-boundary self-damping of the slowest mode pair $(N-1, N)$ in $\widehat{W}_{1,2}$ $(\sim \mathcal{O}(t^{-7/2}(\ln t)^{-5}))$ compared with $\widehat{W}_{0,1}$ across $t \in [10^3, 10^8]$. **(d)** Transverse scaling of order-M Hurwitz reproducing kernels $K_{\boldsymbol{\alpha}}(t_0 - \mathrm{i}d, t_0 - \mathrm{i}d)$ at a multiplicity-3 coalescence point $(m = 3)$, showing $\propto d^4$ for $M = 1$, $\propto d^2$ for $M = 2$, and the strictly positive order-d^0 floor $+\alpha_2\alpha_3(\Xi'''(t_0))^2 > 0$ for $M = 3$.

grows as $\frac{1}{80}(\ln\frac{t}{2\pi})^5$ while the slowest-oscillating saddle-horizon modes near $n \approx N(t)$ experience an extra $\omega_m(t)\omega_n(t) = \mathcal{O}(N(t)^{-2})$ suppression because $\omega_n(t) \to 0$ at the horizon. Conversely, for small indices $m, n \ll N(t)$ where $\omega_n(t) \asymp \frac{1}{2}\ln\frac{t}{2\pi}$, both the diagonal terms and the low-index off-diagonal terms scale with the same $(\ln t)^4$ weight, which is precisely where the exponential prime-gauge sieve $\mathcal{E}_X(z)$ of Paper IV and Paper VI [17, 19] annihilates all low-index cross-terms $2 \le n \le X$. Controlling the remaining unaveraged interior cross-terms on $(X, N(t)]$ as $t \to \infty$ without time-averaging remains an open analytical problem.

APPENDIX A. EXPLICIT ANALYTICAL DERIVATIONS AND INTERVAL-ARITHMETIC PROTOCOLS

A.1. **Exact Cauchy–Riemann derivation of the 4D radial Laplace identity.** Let $X_k(w) := \Xi^{(k)}(t - \mathrm{i}d)$ for $k = 0, 1, 2, 3$. Since $\partial_t X_k = X_{k+1}$, $\partial_t \overline{X_k} = \overline{X_{k+1}}$, $\partial_d X_k = -\mathrm{i}X_{k+1}$, and $\partial_d \overline{X_k} =$

$+\mathrm{i}\overline{X_{k+1}}$, differentiating $\overline{X_0(w)}X_1(w)$ twice with respect to d and t yields

$$\begin{aligned}\partial_d(\overline{X_0}X_1) &= \mathrm{i}\big(|X_1|^2 - \overline{X_0}X_2\big),\\ \partial_d^2(\overline{X_0}X_1) &= -\overline{X_2}X_1 + 2\,\overline{X_1}X_2 - \overline{X_0}X_3,\\ \partial_t^2(\overline{X_0}X_1) &= +\overline{X_2}X_1 + 2\,\overline{X_1}X_2 + \overline{X_0}X_3.\end{aligned} \tag{A.1}$$

Adding the second and third lines of (A.1) cancels $\mp\overline{X_2}X_1$ and $\mp\overline{X_0}X_3$ identically, giving $(\partial_d^2 + \partial_t^2)(\overline{X_0}X_1) \equiv 4\,\overline{X_1}X_2$. Taking imaginary parts yields $(\partial_d^2+\partial_t^2)W_{0,1}(d,t) \equiv 4\,W_{1,2}(d,t)$, which upon substituting $W_{0,1} = d\,\widetilde{W}_{0,1}$ and $W_{1,2} = d\,\widetilde{W}_{1,2}$ proves (4.1).

A.2. **Uniqueness of degree $M = 2$ among higher-order Hurwitz kernels.** In the general degree-M Bézout–Wronskian expansion (4.7),

$$\pi\,K_{\boldsymbol{\alpha}}(t-\mathrm{i}d,\,t-\mathrm{i}d) = \sum_{j=0}^{\lfloor M/2\rfloor}\sum_{k=0}^{\lfloor (M-1)/2\rfloor}(-1)^{j+k}\alpha_{2j}\alpha_{2k+1}\,\frac{\mathrm{Im}\big(\overline{\Xi^{(2j)}(w)}\,\Xi^{(2k+1)}(w)\big)}{d},$$

a summand (j,k) is an adjacent Lucas–Jensen Wronskian $\widetilde{W}_{r,r+1}(d,t) = \frac{1}{2d}\partial_d|\Xi^{(r)}(w)|^2$ if and only if $|2j-(2k+1)| = 1$. For $M=1$, $(j,k)\in\{(0,0)\}$ gives $|0-1|=1$. For $M=2$, $j\in\{0,1\}$ and $k\in\{0\}$ give $|2j-1|=1$ for both $(0,0)$ and $(1,0)$, so every summand is an exact adjacent Wronskian ($\widetilde{W}_{0,1}$ and $\widetilde{W}_{1,2}$). For any $M\geq 3$, the pair $(j,k)=(0,1)$ has $|2j-(2k+1)|=3$, introducing the non-adjacent pairing $d^{-1}\mathrm{Im}(\overline{\Xi(w)}\Xi'''(w))$. Thus $M=2$ is the unique polynomial degree above $M=1$ whose reproducing kernel splits purely into adjacent transverse Wronskians $\frac{1}{2d}\partial_d|\Xi^{(r)}(w)|^2$.

A.3. **Directed-rounding interval-arithmetic protocols.** In the standalone verification suite `scripts/verify_paper13_second_order_debranges.py`, all critical-line interlocking bounds in Table 2 are certified using 35-decimal directed-rounding interval arithmetic (`mpmath.iv.dps = 35`): for each γ_k,τ_k,ν_k $(1\leq k\leq 10)$, the derivative jet $\Xi^{(0)}(t),\dots,\Xi^{(4)}(t)$ is enclosed in rigorous intervals accounting for quadrature and truncation bounds, certifying $c_1(\boldsymbol{t}).a>0$, $c_1^{(1)}(\boldsymbol{t}).a>0$, and $\pi K_{1,1/4}(\boldsymbol{t},\boldsymbol{t}).a>0$ across all 30 critical points.

Use of AI and computational verification

In accordance with editorial policies on transparency in scientific computing, the author discloses that the AI research assistant **Gemini** (Google DeepMind) was used to assist in structuring the manuscript, symbolically cross-checking the Hermitian polarization and radial Laplace identities, and implementing the standalone verification suite `scripts/verify_paper13_second_order_debranges.py`. All mathematical proofs, interval enclosures, and 60-decimal/550-decimal numerical claims were independently verified by executing the verification script (exit code 0).

References

[1] J. B. Conrey and X.-J. Li, *A note on some positivity conditions related to zeta and L-functions*, Internat. Math. Res. Notices **2000** (2000), no. 18, 929–940.

[2] G. Csordas, T. S. Norfolk, and R. S. Varga, *The Riemann hypothesis and the Turán inequalities*, Trans. Amer. Math. Soc. **296** (1986), no. 2, 521–541.

[3] L. de Branges, *Hilbert Spaces of Entire Functions*, Prentice-Hall, Englewood Cliffs, NJ, 1968.

[4] L. de Branges, *The Riemann hypothesis for Hilbert spaces of entire functions*, Bull. Amer. Math. Soc. (N.S.) **15** (1986), no. 1, 1–17.

[5] L. de Branges, *The convergence of Euler products*, J. Funct. Anal. **107** (1992), no. 1, 122–210.

[6] W. Gabcke, *Neue Herleitung und explizite Restabschätzung der Riemann-Siegel-Formel*, Ph.D. dissertation, Georg-August-Universität Göttingen, 1979.

[7] M. Griffin, K. Ono, L. Rolen, and D. Zagier, *Jensen polynomials for the Riemann zeta function and other sequences*, Proc. Natl. Acad. Sci. USA **116** (2019), no. 23, 11103–11110.
[8] A. Hinkkanen, *On functions of the form* $\sum_{n=1}^{\infty} a_n n^{-s}$, Ann. Acad. Sci. Fenn. Math. **22** (1997), 215–226.
[9] R. L. Hudson, *When is the Wigner quasi-probability density non-negative?*, Rep. Math. Phys. **6** (1974), no. 2, 249–252.
[10] H. Ki, *The zeros of the derivative of the Riemann ξ-function*, Proc. Amer. Math. Soc. **136** (2008), no. 9, 3245–3252; see also Adv. Math. **217** (2008), 1599–1621.
[11] J. C. Lagarias, *On a positivity property of the Riemann ξ-function*, Acta Arith. **89** (1999), no. 3, 217–234.
[12] J. C. Lagarias, *Hilbert spaces of entire functions and Dirichlet L-functions*, in *Frontiers in Number Theory, Physics, and Geometry I*, Springer, Berlin, 2006, pp. 365–377; see also Moscow Math. J. **6** (2006), no. 2, 321–346.
[13] D. H. Lehmer, *Extended computation of the Riemann zeta-function*, Mathematika **3** (1956), 102–108.
[14] T. Missiaen, *Complex Saddle-Point Coalescence and Wigner Phase-Space Positivity for the Transverse Wronskian of Riemann's Ξ-Function*, Preprint, Paper I of the Transverse Wronskian Series, 2026.
[15] T. Missiaen, *Parity-Reduced Wigner Kernels, Exact Recurrence Moment Laws, and Transverse Stiffness for Riemann's Ξ-Function*, Preprint, Paper II of the Transverse Wronskian Series, 2026.
[16] T. Missiaen, *Harmonic Chladni Nodal Chambers, Subharmonic Maximum Principles, and Monotonic Crest-to-Zero Flux of Riemann's Ξ-Function*, Preprint, Paper III of the Transverse Wronskian Series, 2026.
[17] T. Missiaen, *De Branges Reproducing Kernels, the Universal Transverse Square-Cascade for Lehmer Pairs, and the Leibniz–Wronskian Identity for Riemann's Ξ-Function*, Preprint, Paper IV of the Transverse Wronskian Series, 2026.
[18] T. Missiaen, *Universal Mother-Wavelet Decomposition of the Jacobi–Riemann Kernel, Smooth Jacobi–Sinh Shells, and Explicit Saddle-Point Remainder Bounds for Riemann's Ξ-Function*, Preprint, Paper V of the Transverse Wronskian Series, 2026.
[19] T. Missiaen, *The Sieved Mirror Kernel $\Phi_X(u)$, the Zero-Free Exponential Prime-Gauge Factor $\mathcal{E}_X(z)$, and Quadratic Prime-Power Purity on $(X, X^2]$ for Riemann's Ξ-Function*, Preprint, Paper VI of the Transverse Wronskian Series, 2026.
[20] T. Missiaen, *Mirror-Paired Hadamard Kernels, Variational Monotonicity Laws, and Explicit Krein-Lobe Bounds for Riemann's Ξ-Function*, Preprint, Paper VII of the Transverse Wronskian Series, 2026.
[21] T. Missiaen, *Mellin–Barnes Spectral Bridges, Montgomery–Vaughan Off-Diagonal Damping, and Preissmann Gram-Matrix Positivity for Riemann's Ξ-Function*, Preprint, Paper VIII of the Transverse Wronskian Series, 2026.
[22] T. Missiaen, *The Three-Part u-Domain Bridge Identity, Phase-Independent Transverse Increment Ellipses, and Analytical Stieltjes–Rosser–Schoenfeld Envelopes for Riemann's Ξ-Function*, Preprint, Paper IX of the Transverse Wronskian Series, 2026.
[23] T. Missiaen, *Beyond the Bohr–Kronecker Torus Barrier: Semiprime Phase-Locking, Quantized Hadamard–Krein Flux Defects $\pi\,\Theta(d - d_*)$, and the Exact Chladni–Wronskian Wall Identity for Riemann's Ξ-Function*, Preprint, Paper X of the Transverse Wronskian Series, 2026.
[24] G. Pólya, *Bemerkung über die Integraldarstellung der Riemannschen ξ-Funktion*, Acta Math. **48** (1926), 305–317.
[25] B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsber. Königl. Preuss. Akad. Wiss. Berlin (1859), 671–680.
[26] C. L. Siegel, *Über Riemanns Nachlaß zur analytischen Zahlentheorie*, Quellen Stud. Gesch. Math. Astron. Phys. Abt. B **2** (1932), 45–80.
[27] A. Speiser, *Geometrisches zur Riemannschen Zetafunktion*, Math. Ann. **110** (1934), no. 1, 514–521.
[28] M. Suzuki, *Positivity of certain functions associated with analysis on elliptic surfaces*, J. Number Theory **131** (2011), no. 10, 1770–1796.
[29] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford Univ. Press, Oxford, 1986.
[30] E. P. Wigner, *On the quantum correction for thermodynamic equilibrium*, Phys. Rev. **40** (1932), no. 5, 749–759.

Independent Researcher, Belgium
Email address: thijs.missiaen@gmail.com